Transportation Research Part E 180 (2023) 103332



Contents lists available at ScienceDirect

Transportation Research Part E

journal homepage: www.elsevier.com/locate/tre





Carbon peak simulation and peak pathway analysis for hub-and-spoke container intermodal network

Xiaoyan Guo [a], Junliang He [a,b,*], Hang Yu [b], Mei Liu [b]

[a] Institutes of Logistics Science and Engineering, Shanghai Maritime University, Shanghai, China
[b] China Institute of FTZ Supply Chain, Shanghai Maritime University, Shanghai, China

ARTICLE INFO

Keywords:
Carbon peak of intermodal networks
Carbon peak simulation
Transport structuring
Network connectivity
Low carbon transport

ABSTRACT

The global consensus on carbon peak and carbon neutral goals extends to the transport sector. This paper provides a method for simulating the carbon emission peak time and peak value within a container intermodal network (CIN). Our approach commences by capturing the fundamental network elements, including nodes, routes, and freight flows, through intermodal network modeling. We then integrate a carbon emission model with scenario analysis, utilizing Monte Carlo simulation to predict the probability of the CIN's carbon emission peak time and value. Finally, using the Yangtze River Delta regional container intermodal network (YRD-CIN) as a case study, 15 scenarios are set up to explore the effects of intermodal transport network development on carbon peaks and peak times under varying conditions in the YRD region. The findings demonstrate the significant influence of modal transport shifts, network connectivity, and energy technology levels on the carbon peak in the YRD-CIN.

1. Introduction

Climate warming caused by greenhouse gas (GHG) emissions has been a significant political and economic concern for over 40 years. The global consensus on carbon reduction has been established since the Paris Agreement of 2015. Over 44 countries and regions worldwide have formally announced their carbon neutrality targets through national legislation or policy announcements by 2020. Transportation, while essential for promoting connectivity, trade, and economic growth in an increasingly globalized world, also contributes significantly to GHG emissions. In 2020, the global transport sector emitted 8,258 million tons of CO_2, accounting for 24 % of the total global emissions, ranking the second carbon emission globally (UN, 2021). The freight sector alone accounts for over 33 % of transport emissions. In China, transportation carbon emissions in 2019 reached 1,274 million tons, ranking second to the United States and accounting for 14.82 % of the global transportation carbon emissions (Peining Tian et al., 2023). Therefore, sustainable transportation plays an essential role in achieving global sustainable development goals (UN, 2021), and various countries have high expectations for future carbon reduction efforts in the transport sector. The European Union has introduced a series of carbon reduction targets since 2020. These targets range from the clean energy application of transport equipment to the optimization of transport networks. For example, by 2030, there will be at least 30 million zero-emission vehicles on the road and zero-emission ships put online, and by 2050, transport will achieve a 90 % reduction target. Moreover, it aims to double the rail freight in 2050 and run the Trans-European Intermodal Transport Network (European Parliamentary, 2021). The U.S. Department of Transportation aims to

* Corresponding author at: 1550 Haigang Avenue, Institutes of Logistics Science and Engineering, Shanghai Maritime University, Shanghai 201306, China.
E-mail address: jlhe@shmtu.edu.cn (J. He).

https://doi.org/10.1016/j.tre.2023.103332
Received 19 July 2023; Received in revised form 18 October 2023; Accepted 19 October 2023
Available online 1 November 2023
1366-5545/© 2023 Elsevier Ltd. All rights reserved.

reduce transportation GHG emissions to 50 %-52 % of 2005 levels by 2030 and 80 %-100 % by 2050 (Muratori et al., 2023). Furthermore, the Chinese government proposes a 9.5 % reduction in transportation carbon emissions intensity compared to 2020, a 40 % share of new energy transportation. The government commits to achieving an oil consumption peak in road transport by 2030 (The PRC, 2021a).

The strategy for reducing carbon emissions in transportation involves revolutionizing energy technologies in vehicles and optimizing transportation resources. Road transport is the primary contributor to transportation carbon emissions, accounting for more than 60 % of the transport sector (Kurtuluş and Çetin, 2020). To address the high carbon emissions associated with road transport, many countries are considering shifting to alternative modes of transportation to alleviate congestion and reduce carbon emissions (Ramalho and Santos, 2021). The European Union's 2050 Transport Route Plan aims to transfer 30 % of road transport to rail or water transport by 2030 and 50 % by 2050 (European Commission, 2011). Similarly, China's Green Transport Plan sets a target of 15 % annual growth in container rail-water transport by 2025 (The PRC, 2021b). As a result, the focus on reducing transport carbon emissions has expanded from a single transport mode to intermodal transport (Guo et al., 2022). Unlike single transport modes, intermodal transport is a holistic transportation pattern of activities that integrates various modes of transportation and facilitates seamless connections. By utilizing transit hubs, freight collection and distribution needs of different nodes can be combined with the provision of transportation resources. Therefore, to achieve the carbon reduction targets for the transport network, it is crucial for the transport sector to evaluate the effectiveness of various carbon reduction measures from an intermodal network perspective.

For the above reasons, this paper presents an innovative methodology for simulating the carbon peak time and peak value within container intermodal networks (CIN). This methodology consists of two main components: modeling container intermodal networks and simulating carbon peaks. The former includes hub identification and freight flow estimation, while the latter entails carbon emission estimation and carbon peak simulation. By employing this methodology, we can illustrate the trends in carbon peak and timing across various scenarios. The findings of this study establish a quantifiable theoretical framework for estimating carbon emissions in intermodal networks, providing a valuable quantitative tool for policy development and assessing the effectiveness of carbon reduction measures in the transportation sector.

The rest of this paper is organized as follows: Section 2 provides an overview of previous research on carbon emissions and peak in transportation, highlighting the existing research gaps. Section 3 presents the research framework and detailed research methods. Section 4 presents the numerical experiment design and data, focusing on a case study of the YRD-CIN. In Section 5, historical carbon emissions of the YRD-CIN are estimated, and the carbon peak and the development trend of the YRD-CIN under different scenarios are predicted and discussed. Finally, Section 6 summarizes the work and highlights the innovations presented throughout the paper.

2. Literature review

Intermodal transport involves comprehensive processes that perform the transportation of goods from their origins to destinations using at least two different modes of transportation (SteadieSeifi et al., 2014). With the widespread adoption of container standardization, container transport has become the primary focus of intermodal transport studies (Basallo-Triana et al., 2021). Container intermodal networks consist of nodes, hubs, channels, and different modes of transport to facilitate the movement of containers between various pairs of nodes (Guo et al., 2022; Talley and Ng, 2018). It enables customized door-to-door services by integrating various modes of transport. Additionally, it leverages the distribution function of hub nodes to achieve economies of scale in logistics and integrates freight distribution demand and transportation resource supply across different nodes (Bontekoning et al., 2004). The study of carbon emission peaks in container intermodal networks facilitates the achievement of sustainable transportation. This paper aims to review previous research findings from two key areas: carbon emission estimation and the exploration of carbon peaks and pathways in transportation networks.

2.1. Carbon emission estimation study in transport

The existing studies on transport carbon emission estimation have made significant progress in two main areas: developing estimation methods and analyzing carbon emission scenarios of transport networks at various scales. These estimation methods can be broadly categorized into top-down models (Peining Tian et al., 2023; Xiong et al., 2023; Zhang et al., 2023) and bottom-up models (Chocholac et al., 2021; Guo et al., 2022; Ramalho and Santos, 2021), while some studies also used mixed models (Kirschstein and Meisel, 2015; Shafiei et al., 2013). Estimation studies of transport carbon emissions are built upon transport network models, and the choices of estimation methods are related to the spatial and temporal scales of the transport networks (Linton et al., 2015). The top-down approach relies on macroscopic energy statistics. WANG et al. (2019) employed the approach to estimate the carbon emissions of five transport modes in three northeastern provinces of China, focusing on fossil fuel consumption. The bottom-up approach includes the development of microscopic transport activity models (IPCC, 2006). Cariou et al. (2019) employed the approach to examine the shipping carbon emissions along global container supply chains by tracking liner services. Additionally, Kirschstein and Meisel (2015) established a hybrid model to capture the characteristics of the study object. This approach involves a microscopic model to improve accuracy and also collects integrated data by a macroscopic approach to overcome data limitations. Linton et al. (2015) conducted a comprehensive review of transport carbon emission estimation methods, ranging from microsimulation to global techno-economic models. The study emphasizes that comprehensive transport emission estimation models can provide a deeper understanding of transport carbon emission issues.

The research on transport carbon emission can be categorized into two distinct dimensions: single transport mode and intermodal transport. The existing literature encompasses a wide range of research on carbon emissions from various transport modes, including

ships (Aakko-Saksa et al., 2023; Wang et al., 2022), rails (Chocholac et al., 2021; Li and Zhang, 2020), roads (Haywood and Jakob, 2023; Linton et al., 2015), and air transport. Studies on a single transport mode carbon emissions have delved into a multitude of facets, such as fuel utilization (Ampah et al., 2021; Lee et al., 2018), infrastructure development (Aakko-Saksa et al., 2023), estimation of emission factors (Franco et al., 2013; Raparthi and Phuleria, 2022; Zhou et al., 2022), identification of impact factor (Wang et al., 2020a; Zhou et al., 2021), evaluation of the policies (Haywood and Jakob, 2023), and other pertinent aspects. These studies have injected comprehensive insights into the carbon emission characteristics of different transport modes. However, in practice, except for road transport, the assessment of carbon emissions for other single transport modes has failed to capture the entirety of the origin–destination freight emissions process.

In contrast to studies focusing on a single mode of transportation, intermodal carbon emission studies examine emissions throughout the entire transportation process from the origin to the destination. Existing research on intermodal transport has primarily demonstrated the carbon reduction effects of intermodal transport chains and analyzed emission efficiency. Kirschstein and Meisel (2015), de Miranda Pinto et al. (2018), Lin (2019) and Tao and Wu (2021) found that the rail-based intermodal transport had lower emissions than those from road transport. Additionally, Kim and Van Wee (2014) and Ramalho and Santos (2021) explored the carbon reduction potential of intermodal transport, particularly in short-sea transport. However, these studies mainly focus on individual intermodal chains and do not extend their analysis to the network level. Heinold and Meisel (2018) attempted to assess the efficiency of intermodal transport emissions by constructing an intermodal transport network model. However, this study uses fictitious freight volume data rather than real traffic flow. As a result, it fails to comprehensively evaluate the overall emission situation of the intermodal network. Guo et al. (2022) estimated intermodal carbon emissions at the network level to address this gap. While this study provides valuable insights, it was limited to a single hub and has not extended its analysis to the network level. To support decision-making in the transportation sector, it is crucial to have a comprehensive understanding of the carbon emission scenarios at the network level. Therefore, it is essential for an intermodal network carbon emissions model to simulate cargo flows and calculate carbon footprints in regional intermodal networks.

2.2. Carbon peak research in transport

Studies on transport carbon peak usually combine transport carbon emission models with scenario analysis to simulate future carbon emission trends. Hao et al. (2015) and Wen and Song (2022) discussed the potential for China's freight system to achieve the carbon peak. Hao et al. (2015) established scenarios including freight demands, energy intensity, technological penetration, and carbon emission intensity. The results show that the carbon peak will be achieved in 2045 under the current scenario and in 2035 under the optimistic scenario. On the other hand, Wen and Song (2022) used a system dynamics model and set multiple scenarios based on rail freight share, rail electrification level, and electric truck penetration. Their results suggest that the Chinese freight system could peak as early as 2024 in the most optimistic scenario or as late as 2029 in the most pessimistic scenario.

Other scholars studied the road network carbon peak. Dong et al. (2022) focused on the potential for road transport to reach the carbon peak target. They considered vehicle composition, fuel economy, and fuel emission factors. Their results indicate that, under the current policy environment, road transport could reach the carbon peak by 2033 regarding fuel composition and fuel economy. Xiong et al. (2023) applied a Monte Carlo simulation and considered the road electrification rate as the scenario parameter. The results show that road transport could peak in 2028 without policy intervention, and increasing the road transport electrification rate is the key to achieving the peak target.

These studies highlight the variations in carbon peak predictions within and across different transport networks. Different scenario settings and network characteristics can lead to different prediction conclusions of carbon peak. Therefore, it is critical to present the characteristics of intermodal networks and design relative scenarios to achieve peak targets for the entire transportation system. However, there is a research gap regarding carbon peaks in intermodal transport networks. Existing research primarily focuses on macro-level analysis, such as the freight, highway, and ship transport sectors. Research on intermodal networks from a regional perspective and carbon peak and peak paths in these networks is seldom performed.

To bridge these gaps, our study focuses on the carbon emissions on an intermodal network at the regional level to investigate the simulation of carbon peak and peak pathways in the CIN. We incorporate the structural attributes of intermodal transport networks into both the modeling of the network and the simulation of carbon peaking. In the scenario analyses, we focus on the impacts of intermodal network construction on carbon peak, peak time, and peak path under different energy environments. The findings in this work provide a valuable quantitative tool for evaluating and formulating effective carbon emission reduction measures and strategic planning in the transport sector.

3. Modeling of carbon peak prediction for hub-and-spoke intermodal networks

3.1. Methodological framework

Modeling the carbon peak of intermodal networks relies on a well-defined physical structure of the network and the logistics data. So, a hub-and-spoke network is adopted to present the intermodal transportation network. To model the framework of the hub-and-spoke intermodal networks, a machine learning-based model is developed to identify hubs and spokes. A freight flow estimation model is developed to estimate the freight flows of intermodal routes and the throughput of its critical hubs. During the hub identification and flow estimation processes, the critical network indicators are developed for accuracy inspection. In the second, for simulating carbon peak, we develop the comprehensive carbon estimation method based on previous emission studies and design the scenario analysis

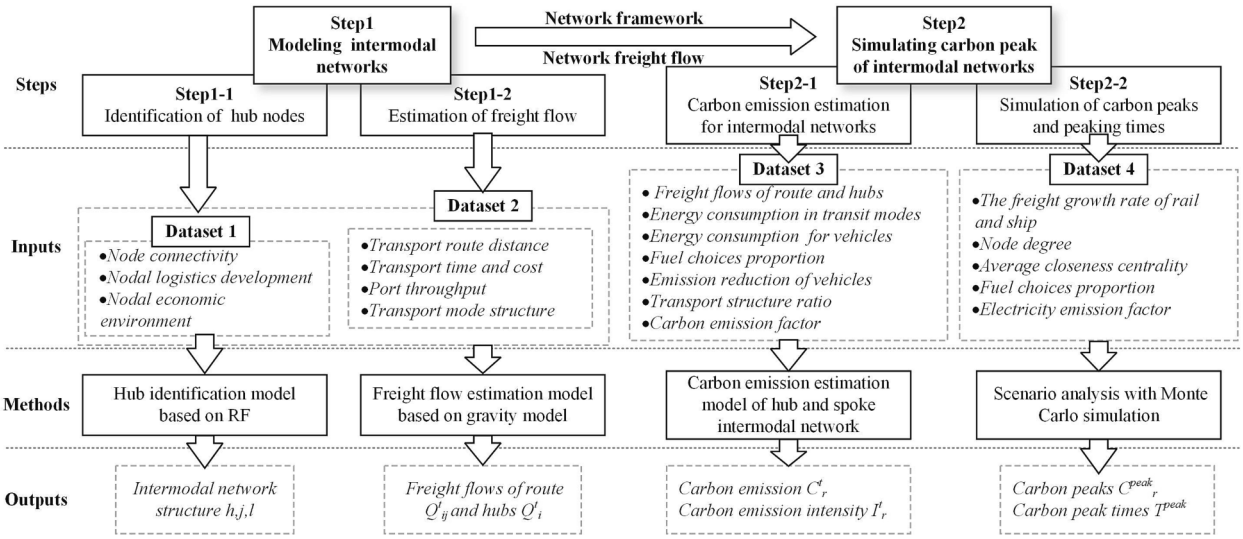


Fig. 1. Methodological framework.

Table 1
Definitions of sets.

Set	Definitions
I	The set of the intermodal transport nodes, $i,j,h \in I$;
H	The set of hub nodes h, $h \in H \subset I$;
K	The set of transport mode k, $k \in K = \{railway, road, inlandwaterway\}$;
A	The set of fuel a used in transit and transportation processes, $a \in A$;
P	The set of transfer operation phases $p \in \{0,1\}$, when $p = 0$ is for cargo in transport,$p = 1$ is for cargo in transit at the node, $p \in P$;
T	The set of periods t, $t \in T$;
B	The methods of emission assessment $\beta \in \{0,1\}$, When $\beta = 0$, carbon emissions are assessed as Well-to- Wheel, and when $\beta = 1$, as Tank- to- Wheel;
L	The set of transport routes l, $l \in L$.

with Monte Carlo simulation. Then, our study provides insights into the trends of carbon emission peaks and peak times within the intermodal networks under different scenarios. The methodological framework is shown in Fig. 1.

The symbolic interpretation of the sets and parameters in the method is displayed in Tables 1 and 2.

3.2. Modeling of hub-and-spoke intermodal networks

3.2.1. The key elements and assumptions of hub-and-spoke intermodal networks

Hub-and-spoke intermodal networks offer a valuable framework for understanding the structure, function, and characteristics of intermodal networks (Meng and Wang, 2011). These networks demonstrate the advantages of economies of scale and strategic location in container transport (Basallo-Triana et al., 2021). In this study, we utilize a hub-and-spoke network structure (As shown in Fig. 2) to describe the intermodal network, which comprises the following key elements:

(1) Hub nodes h: These are intermodal transfer service providers within the network and serve as significant collection and distribution centers for containers. Our study focuses on hub nodes that facilitate intermodal transfers across three transportation modes of road, rail, and ship. A regional container intermodal network consists of several hub-centered intermodal subnetworks.

(2) Spoke nodes j: These nodes represent the local cargo delivery and arrival locations within the hub-centric intermodal subnetwork.

(3) Routes (links)l: These are the interconnected container transport routes between nodes, utilizing rail, ship, and truck.

(4) Container cargo flows and directions Q_{ij}: The containers move between nodes, categorized as outgoing or incoming volumes. It reflects the demand for intermodal container transport between different nodes.

To simplify the complexity of intermodal networks, our study incorporates the following assumptions:

(1) Regional hub exerts varying degrees of influence on other regional nodes over a specific period. The intensity of this influence is determined by the logistics attractiveness of each node.

(2) The number of container transfers will not exceed two times.

(3) Feeder nodes are excluded from the node selection process. But, the carbon emissions of the feeder nodes are still taken into account. We assume the presence of feeder distribution areas primarily served by road transportation at the selected intermodal nodes. The distance between the intermodal nodes and feeder nodes is assumed to be the average road distance from the nodes to the city

Table 2
Parameter interpretation.

Parameter	Definitions of the hub identification model
s_i^k	Node degree on transport mode k network of node i;
x_{ij}^k	Adjacency matrix of the network, if there is a direct edge between node i to node j in transport mode k network $x_{ij}^k = 1$, otherwise $x_{ij}^k = 0$;
$C_c^k(i)$	Closeness centrality on transport mode k network of node i;
n^k	Number of intermodal transport nodes in the network of transport mode k;
d_{ij}^k	Shortest distance from node i to node j by the transport mode k;
δ_{ij}^k	If the container is transported from node i to node j using transport mode k,$\delta_{ij}^k = 1$, otherwise $\delta_{ij}^k = 0$;
GE^k	Network efficiency of the network of transport mode k;
Q_{ij}^k	Container freight volume from node i to node j by the transport mode k;
w_{ij}^k	Composite link weights from node i to node j by the transport mode k;
N	Total number of intermodal nodes;
X_i	Feature Value,$X_n \in \mathbb{R}^n$;
y_i	$y_n \in \{0,1\}$,$y_i = 1$ indicates that i is the hub node, otherwise $y_i = 0$.
Parameter	Definitions of the flow estimation model
Q_{ij}^t	Container freight volume from node i to node j in period t;
Q_i^t	Container freight volume of node i in period t;
$f(d_{ij})$	Distance function from node i to node j;
$\tau,\partial,\omega,\rho$	Correction factors;
R_{ij}^t	Logistics attractiveness from node i to node j in period t;
G_i^t	Logistics influence factor of node i in period t;
M_i^t	Relative logistics quality coefficient of node i in period t;
M_i^t	Logistics quality factor of node i in time period t;
$t_{ij}^k, c_{ij}^k, d_{ij}^k$	Transportation time, cost and distance from node i to node j by transport mode k;
$Q_{ij}^{'t}$	Modified freight volume from node i to node j in period t.
Parameter	Definitions of network carbon emission estimation
C_r^t	Total carbon emissions of the CIN r in period t;
$C_h(x_n^t)$	Carbon emissions from the CIN of the hub node h network in period t;
x_n^t	Critical parameters for carbon emission modeling in period t;
$Q_{hjk}^{p,t}$	Container activity indicators, indicating the volume of container freight transported by mode k from hub node h to node j in operation phase p;
$B_{hjk}^{p,t}$	Activity efficiency index, when $p = 0$, indicates that the operation efficiency per hour in the hub node h, container transfer to transport mode k; when $p = 1$, indicating that the transport time from the container hub node h to node j through transport mode k;
$S_{hjk}^{p,t}$	Energy intensity index, indicating the operational energy consumption per hour of fuel a for containers transported from hub node h to node j by transport mode k in operational phase $p(p = 0)$, or the operational energy intensity of hub node h transfer to transport mode $k(p = 1)$;
$E_{hjka}^{p,t}$	Energy composition indicator, indicating the share of energy a in transport activities or transfer activities;
λ_{hj}^t	Transport structure ratio, indicating the share of containers transported by transport mode k at hub node h;
F_a^β	Carbon emission factor of fuel a under carbon emission assessment model β;
I_r^t	Carbon emission intensity of the CIN r in period t;
Q_r^t	Total container volume of the CIN r in period t;
$\overline{d}_r$	Average transport distance of the regional intermodal network r;
C_r^{peak}	Carbon peak value of intermodal network r;
T_r^{peak}	Carbon peak time of intermodal network r.

center.

(4) Different economic distances exist for ship and rail transport within the hub network.

(5) In a hub-centric intermodal subnetwork, the spoke nodes are only connected to the hub nodes, not to the spokes. Thus, the hub-and-spoke network only considers the container volume transferred through the hub nodes.

3.2.2. Critical network indicators

To further explore the impact of network structure on intermodal network emissions, this study integrates the network elements into the intermodal network simulation as well as scenario analysis by constructing a set of indicators. These indicators include the node connectivity indicators and the network efficiency indicator. The node connectivity indicators define the range of services nodes provide. It is a training parameter for the hub identification model. It also plays a significant role in estimating the logistics attractiveness of nodes in the freight flow estimation model. The network efficiency indicator describes various network connectivity scenarios. It is the basis for scenario setting in analyzing network elements.

(1) Node connectivity indicators.

Node connectivity indicators are one of the evaluation dimensions of accessibility and indicate the ease with which nodes in complex networks can reach other nodes (Lü et al., 2016). In this study, the node connectivity indicators are developed considering various transportation modes using node degree and node closeness centrality, as shown in Equation(3.1) and Equation (3.2).

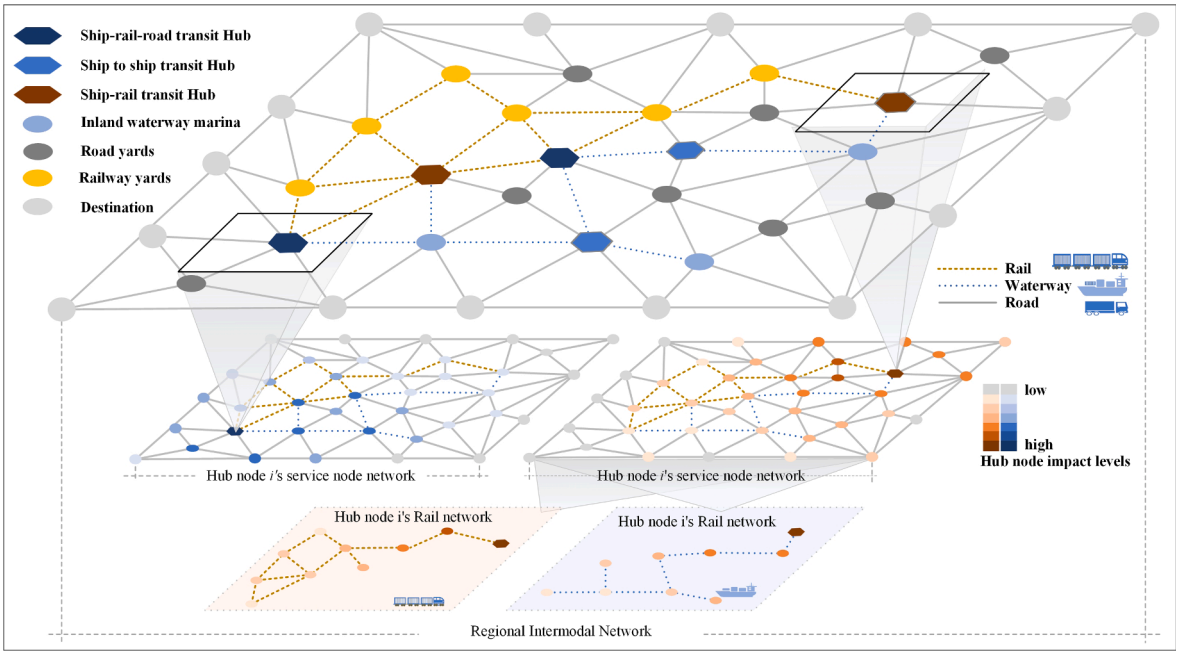


Fig. 2. The structure of intermodal transport network.

$$s_i^k = \sum_j^{n^k} x_{ij}^k, \quad \forall j \neq i \leqslant n^k \in I, k \in K \tag{3.1}$$

$$C_r^k(i) = \frac{n^k - 1}{\sum_{j \neq i \in G}^{n^k} d_{ij}^k}, \quad \forall 1 < j \neq i \leqslant n^k \in I, k \in K \tag{3.2}$$

Equation (3.1) represents the node degree of node i in the transport mode k network. It indicates the number of neighboring nodes directly connected to node i. The larger the s_i^k, the wider the scope of node i services. Equation (3.2) represents the closeness centrality of node i in the transport mode k network, which indicates how conveniently it reaches other nodes.

(2) Network efficiency indicator.

Network efficiency is for a comprehensive assessment of network circulation. It is calculated as the average efficiency between nodes in the network (Latora and Marchiori, 2001). By referring to the modeling of Latora and Marchiori (2001) for network efficiency, we incorporate the route container volumes to establish an intermodal network efficiency model. This model provides a more intuitive reflection of the network efficiency with different container flows across various transportation modes, as shown in Equation (3.3). It comprehensively demonstrates the level of intermodal network efficiency under different transportation structure ratios and node connectivity.

$$GE^k = \frac{1}{N(N-1)} \sum_{i \neq j \in I} \frac{1}{w_{ij}^k}, \quad 1 < N, k \in K \tag{3.3}$$

$$w_{ij}^k = \sum_{k \in K} \delta_{ij}^k \cdot \frac{d_{ij}^k}{Q_{ij}^k} \quad \forall j \neq i \in I \tag{3.4}$$

Equation (3.4) explains that the weighting factor GE^k of the network efficiency GE^k is composed of the shortest path distance d_{ij}^k and the capacity Q_{ij}^k. The network efficiency is inversely proportional to the average path distance but positively proportional to the link capacity.

3.2.3. Hub identification model based on machine learning

Hub nodes are vital in container cargo flows of a hub-and-spoke intermodal network. Identifying the hub nodes quickly and accurately from the vast pool of interconnected nodes in the intermodal network is important. Thus, we develop a machine learning-based model for identifying key hub nodes. Among various machine learning algorithms, random forest (RF) exhibits exceptional performance even with high-latitude data (Cheng et al., 2019; Paul et al., 2018). Additionally, RF has improved generalization ability and stability, as well as it can discover the influence between features in the classification process (Khajavi and Rastgoo, 2023; Rudzianskaite-Kvaraciejiene et al., 2015). Therefore, RF is employed in this study for training to identify the critical hub nodes within

the regional intermodal network. This model is applied within the framework of classical algorithms. Similar applications of traditional methods, rather than algorithmic innovations, are evident in other studies. For instance, in the work of Göçmen and Erol (2019).

In our study, we first construct a dataset $\{(X_i, y_i) : i \in [m] := \{1, 2, \ldots, m\}\}$ for machine learning. Previous research on logistics hub node identification typically incorporates indicators such as the economic environment, logistics infrastructure, and freight volume levels (Zhiqiang Cao et al., 2022). Our study expands the considerations beyond these traditional indicators and includes node connectivity indicators that characterize the network structure.

3.2.4. Modeling freight flow estimation based on the gravity model

Freight transport activities within intermodal networks are a primary carbon emission source. These activities are contingent upon the transfer and transportation of freight between nodes in the network. Consequently, following the construction of the intermodal network framework, it becomes imperative to assess the freight flow between these nodes, considering the freight volume of the hub nodes. It effectively converts the static physical network into a dynamic logistics network. Due to the inherent complexity of intermodal networks, obtaining real Origin-Destination (OD) data requires a substantial investment of time and resources. In cases where the OD flow matrix is challenging to acquire or partially unavailable, the gravity model (GM) offers a straightforward and efficient method for estimating the freight flow between ODs (Tamin and Willumsen, 1989).

The GM assumes that OD freight flows are proportionate to a region's size factor while inversely proportional to its distance function. Subsequent research has expanded the applicability of these functions to a broader range (Arbues and Banos, 2016). Numerous studies on flow estimation have leveraged gravity modeling to gauge the attractiveness of freight allocation and capture the interdependence between destinations and originating transportation activities (Reul et al., 2021). As a result, GM has emerged as a pivotal tool for transportation flow estimation and finds widespread use in estimating freight trips (Kalahasthi et al., 2022), analyzing transportation demand (Tamin and Willumsen, 1989; Yu et al., 2021), studying transportation networks (Wojahn, 2001), and estimating freight flows (Arbues and Banos, 2016; Levine et al., 2009; Shen and Aydin, 2014). Moreover, GM plays a vital role in supporting studies related to transportation emission estimation (Guo et al., 2022; Randrianarisoa and Gillen, 2022; Xu et al., 2021).

Based on prior research, our study introduces the gravity model to estimate freight flows within container intermodal networks, further enhancing the freight flow estimation model. This model relies on assessing the logistics quality of container transportation in node cities to derive an attractiveness ratio, which serves as a pivotal parameter in evaluating node logistics attractiveness and subsequently constructing an improved freight flow estimation model.

$$Q_{ij}^{t} = Q_{i}^{t} \cdot R_{ij}^{t} \cdot \left(\sum_{j} R_{ij}^{t}\right)^{-\gamma}, \quad \forall j \neq i \in I \tag{3.5}$$

$$R_{ij}^{t} = \frac{G_{i}^{t} \cdot M_{i}^{'} \cdot M_{j}^{'}}{f(d_{ij})} \tag{3.6}$$

$$G_{i}^{t} = \frac{M_{j}^{'}}{M_{i}^{'} + M_{j}^{'}} \tag{3.7}$$

$$M_{i}^{'} = \frac{1}{1 + e^{-M_{i}^{t}}} \tag{3.8}$$

$$f(d_{ij}) = \frac{1}{K} \sum_{k \in K} (t_{ij}^{k} \cdot c_{ij}^{k} \cdot d_{ij}^{k}), \quad \forall j \neq i \in I \tag{3.9}$$

Among them, Equation (3.5) represents the model for estimating container freight flow from node i to node j. This model primarily comprises the total freight at node i and the logistics attractiveness index from node i to node j during the period t. The logistics attractiveness shown in Equation (3.6) reflects that the logistics attraction increases with improved logistics quality and decreases with longer integrated distances between cities. G_i^t (Eqn 3.7) consists of the relative logistics quality factor $M_i^{'}$ between nodes. The logistics quality coefficient M_i^t is obtained by constructing the logistics development evaluation index for the nodes in the network in the factor analysis. The greater the logistics quality coefficient of a node, the higher its logistics attractiveness. Equation (3.8) represents a positive treatment for M_i^t. In Equation (3.9), the integrated logistics distance coefficient accounts for the negative effects of transportation cost c_{ij}^k, time t_{ij}^k, and distance d_{ij}^k associated with different transportation modes on the logistics attractiveness index. The coefficient of gravity corrections γ is incorporated into the model to adjust the overall attractiveness based on the gravity principles.

The above processes present the construction of a network freight flow estimation model, however with the absence of OD data. In the practical application process, once some OD data is obtained through enquiries and investigations, the freight volume can be adjusted using Equation (3.10). The method of data refurbishment uses the logistic regression approach illustrated in Shen and Aydin (2014).

$$Q_{ij}^{'} = \tau \cdot (Q_{i}^{t})^{\partial} \cdot (R_{ij}^{t})^{\omega} \cdot f(d_{ij})^{-\rho}, \quad i \neq j \in I \tag{3.10}$$

The estimation of the freight flow of an intermodal network includes two steps. Firstly, the freight volume from the hub to each

spoke node and other hubs is calculated based on the container delivery volume from the hub to the network at a specific time. Secondly, the arrival volume at the hub calculates the freight volume from the spoke node back to the hub. This approach allows for determining container flow and its direction over the entire network, mitigating the risk of double counting on transport paths.

3.3. Methodology for simulating carbon peak in intermodal networks

3.3.1. Modelling of carbon emission estimation of the hub and spoke intermodal network

In our previous study, it was concluded that the carbon emissions of intermodal networks are influenced by several key factors under a given freight volume, i.e., transport structure, transport activity efficiency, network energy structure, and operational activity efficiency of transshipment nodes (Guo et al., 2022). These critical factors are used to develop a carbon emission estimation model for a regional intermodal network based on multiple hubs. The estimation model combines the carbon emission estimation model for a single hub (Guo et al., 2022) with the simplification approach of the activity-based carbon emission measurement model conducted by Tao and Wu (2021). The bottom-up approach is used to model carbon emissions of intermodal networks, while a top-down approach is involved to obtain data on critical parameters of the model. It ensures the accuracy of the model, the availability of data, and the realization of the carbon peak scenarios. The specific details are shown as follows.

$$C_r^t = \sum_{h \in H} C_h(x_n^t), \quad \forall t \in T, h \in H \subset I, n > 0 \tag{3.11}$$

$$C_h(x_n^t) = \sum_{jpka} Q_{hjk}^{p,t} \cdot B_{hjk}^{p,t} \cdot S_{hjk}^{p,t} \cdot E_{hjka}^{p,t} \cdot F_a^{\beta}, \quad h \neq j \in I, k \in K, p \in P, \beta \in B, a \in A \tag{3.12}$$

$$Q_{hjk}^{p,t} = Q_{hj}^t \cdot \lambda_{hj}^t \tag{3.13}$$

Equation (3.11) represents the total carbon emissions of the regional container intermodal transport network that is composed of the sum of the hub node network carbon emissions within the network. Equation (3.12) represents the construction of the intermodal network carbon emission estimation model based on container freight volume flow, intermodal activity efficiency, operational equipment energy intensity, and energy structure from the perspective of intermodal network transportation activities. Equation (3.13) represents the container volume flow of different transport modes at each hub node obtained through its transport structure ratio λ_{hj}^t and freight flow estimation of Q_{hj}^t. Data for the other factors is obtained upon the model in our previous study (Guo et al., 2022).

In addition, based on the regional network carbon emission model, we also define the carbon emission intensity for unit containers in the regional network.

$$I_r^t = C_r^t / Q_r^t / \overline{d}_r \tag{3.14}$$

$$\overline{d}_r = \frac{1}{K} \sum_{k \in K} \frac{2}{n^k (n^k - 1)} \sum_{i \neq j \in I} d_{ij}^k, \quad k > 0, n^k > 1 \tag{3.15}$$

As indicated in Equation (3.14), the network carbon intensity refers to the carbon emissions generated by a container traveling every kilometer within an intermodal transport network. It serves as a measure of the overall carbon emission efficiency of the intermodal network. The average transport distance $\overline{d}_r$, as depicted in Equation (3.15), represents the average path distance of an intermodal network encompassing different transportation modes.

3.3.2. Predicting of carbon peaks and peaking times based on Monte Carlo simulation

The scenario analysis allows for discussing static trends in carbon emission changes by manipulating parameter settings. However, it fails to capture the dynamic processes of carbon peak when faced with parameter uncertainty. Therefore, a Monte Carlo simulation approach is employed in this study to quantify the uncertainty of variables that influence carbon emissions in intermodal networks. This method considers the range of parameter variability in the scenario analysis and simulates changes in carbon emission peaks and peaking times within the regional intermodal network under current governmental policies and future socio-economic development scenarios. The steps of the method are as follows:

(1) The processes of establishing risk variables and constructing probabilistic: The carbon emission estimation model of the intermodal transport network discussed in the previous section is incorporated. The change rates of transport structure λ_{hj}^t, energy structure $E_{hjka}^{p,t}$, activity efficiency $B_{hjk}^{p,t}$ and energy intensity $S_{hjk}^{p,t}$ are considered risk variables. Using historical data, we capture the pre-distribution of their change rates. We assume that the change rates of those variables follow a normal distribution. Through various scenario parameter distributions, the mean $\sigma_{x_n}^t$ and standard deviation $\mu_{x_n}^t$ of the change rates for critical parameters x_n^t are available in future years.
(2) The processes of random sampling and statistics: In the random sampling process, random variables that adhere to the known probability distribution of variables x_n^t are identified. We perform 100,000 random simulations using Crystal Ball software to

Table 3
Comprehensive data of nodal cities.

Indicators		Data	Source
Logistics environment indicators	Economic environment	GDP, fiscal revenue, GDP per capita, etc.	Statistical yearbook data and transportation development bulletin data for 41 cities from 2012 to 2021
	Industry development	The gross value of the secondary industry, gross industrial product, gross postal and telecommunications services, gross express services, etc.	
Freight capacity indicators	Freight volume data	Port throughputs	
	Freight structure	Freight volume structure ratio, rail, road, and water freight volume	
Regional network structure indicators	Node connectivity indicators	Node degree, node closeness centrality	

obtain the probability distribution of each variable. By dynamically transforming each variable at different time points, the carbon emissions of the intermodal network are predicted for different times in the future.

(3) The processes of the prediction of carbon emission peaks and peak times: Using the settings above and simulations, we utilize Crystal Ball to set the future maximum network carbon emission value C_r^{peak} and the corresponding time T_r^{peak} as predictor variables, as depicted in Equation (3.16) and Equation (3.17). The carbon emission peaks and peak times within dynamic intermodal networks under different parameter variations are determined.

$$C_r^{peak} = C_r^{t+n} = \max(C_r^t, C_r^{t+1}, \cdots, C_r^{t+n}, \cdots), n \geqslant 0 \tag{3.16}$$

$$T_r^{peak} = t + n \tag{3.17}$$

(4) The processes of the presentation of carbon emission peak and peak time ranges using probability distribution diagrams: Through the analysis of 100,000 simulations, we present the frequency and probability of different carbon peaks and peaking times using probability distribution diagrams. Additionally, the trends of these peaks and peak times are formed, facilitating a comprehensive discussion on the trends of carbon peak within intermodal transport networks under different parameter scenarios.

4. Numerical experiments design and data

Taking the YRD-CIN as a case study, the implementation process of the aforementioned methodology is illustrated. This section presents an overview of the initial parameter of the YRD-CIN. Based on the initial parameter, several parameters for various scenarios are designed.

4.1. Experimental design

We designed the experiment with five steps based on the methodology presented in Section 3. Firstly, the datasets are established through collection from literature and surveys. Second, critical elements of intermodal networks are captured by node identification models based on machine learning. These elements include the hub and hinterland nodes and the routes linking them to each other within the hub-and-spoke intermodal network. Thirdly, the transit flows in the hub nodes and transport flows along the routes of the intermodal network are calculated according to the freight flow estimation model. Then, these flows would be fed into the emission estimation model as critical elements of the intermodal transport activity indicator, and carbon emissions would be estimated. Finally, a dynamic simulation of carbon peaking in intermodal networks is implemented through scenario design and Monte Carlo simulation based on the critical parameter variations of the carbon emission estimation model. The specific experimental steps are shown in Appendix Table A2.

Our experiments are conducted using various tools. We utilized the Scikit-learn library in Python 3.7.4 for training the machine learning model, SPSS for conducting logistics quality coefficient evaluations, Gephi for calculating network metrics, Excel built-in functions for implementing carbon emissions calculations, and Oracle Crystal Ball 11.1.41 for simulating carbon peak. These experiments are done on computations configured with a processor of 12th Gen Intel(R) Core (TM) i7-12700H, with a 32 GB RAM.

4.2. Initial parameter

4.2.1. Intermodal network modelling dataset

The modeling intermodal network datasets consist of the hub identification and freight flow estimation datasets. The Hub identification dataset includes comprehensive node city and regional transport network data. The city node data comprises the national statistical yearbook data from 41 cities for evaluating hub nodes and estimating traffic. Examples of logistics environment indicators,

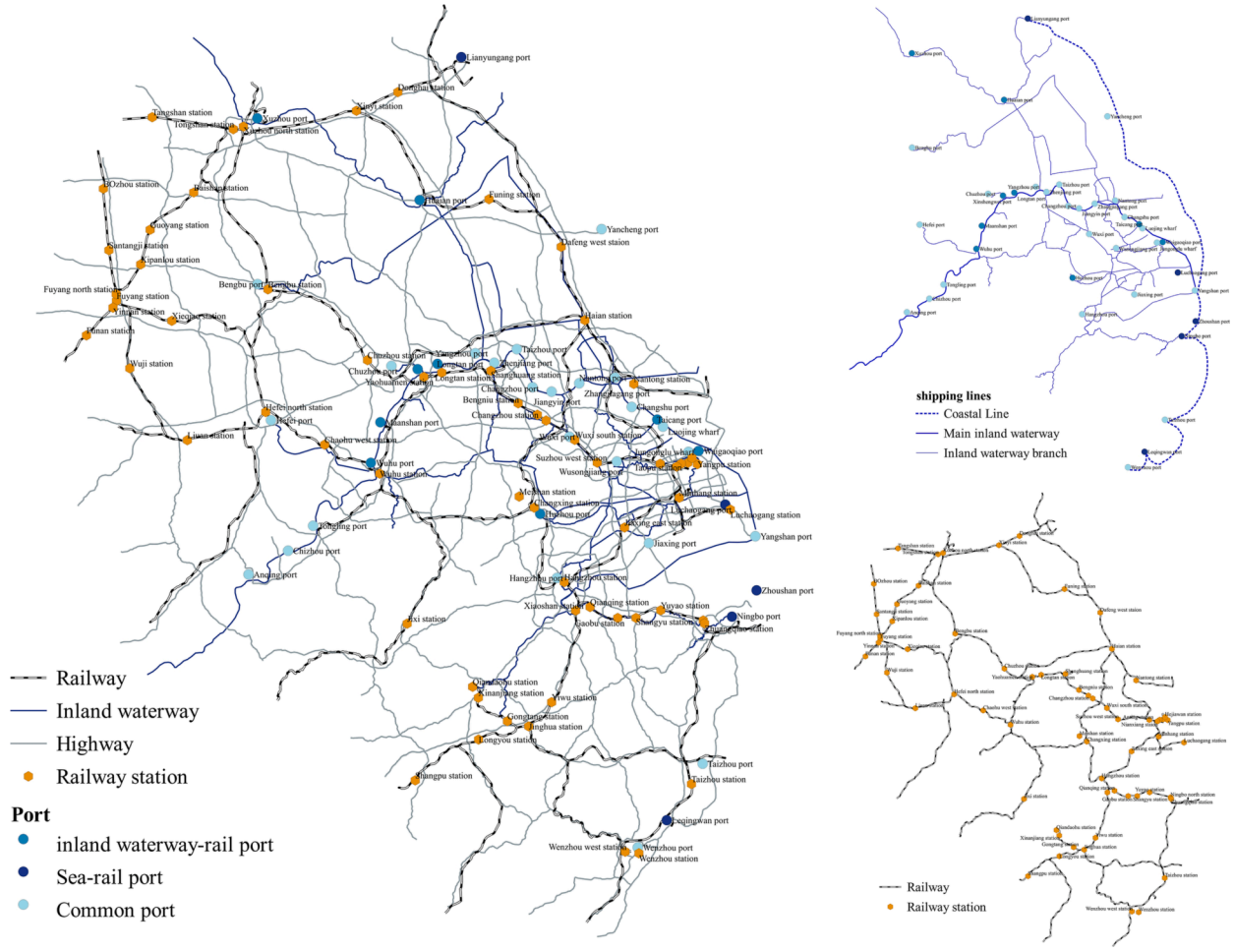


Fig. 3. The YRD intermodal transportation network.

Table 4
The energy intensity of different transit modes.

Type of transit mode	Energy intensity (MJ/TEU)
Ship-rail	(167.7–527.7)
Sea-inland waterway	(135.1–214.6)
Rail-road	(97.2–154.9)
Waterway-road	(102.1–127.8)

freight capacity indicators, and network structure indicators for different nodes are presented in Table 3. In Table 3, detailed data is available on the websites of the provincial statistical offices. The label data for machine learning are determined based on the relative logistics quality coefficient (M_i^t). As the hub node, we identify cities with the M_i^t greater than 0.5. This method was used by Asgharian Rezaei et al. (2023).

The regional transport network data abstracted from the case of the YRD-CIN includes rail networks, inland waterway networks, and integrated transportation networks, as illustrated in Fig. 3. Distance data for different transport modes are actual distances captured via the Baidu Maps API.

Furthermore, we estimate the container throughput of the hub nodes serving the YRD-CIN using the hub container throughput data from the previous decade using the flow estimation model proposed in this paper. Future freight volumes are modeled using a regression approach to facilitate the simulation of carbon peak in the YRD-CIN.

4.2.2. Carbon emission estimation modelling dataset

The carbon emission estimation dataset includes energy consumption data of the operational processes in the regional intermodal network. Those data encompass the transit process and transportation process. The energy consumption data of different transit modes

Table 5
Energy consumption for different transport vehicles.

Mode	Parameter type	Selected		Parameter interval	Sources
Ships	Vessel type	Feeder vessels	Trunk vessels		
	Maximum capacity (TEU)	64	124	44–208	(Fan et al., 2023, 2021a; Shu et al., 2023; Sun et al., 2013; Yan et al., 2023)
	Engine Power (kW)	140*2	360*2	214.6–1384	
	Service speed (kt)	10.5	18	8–25	
	Engine specific energy consumption per hour (g/kWh)	215	175	166–250	
Trains	Electricity consumption (kWh/TEU-km)	0.219		0.169–0.329	(Chocholac et al., 2021; de Miranda Pinto et al., 2018; Heinold and Meisel, 2018; Huang et al., 2021; Li and Zhang, 2020; Lin, 2019; Sun et al., 2022; Wang and Zhu, 2019)
	Oil consumption (MJ/TEU-km)	2.98		1.83–4.62	
Trucks	Oil consumption (L/TEU-km)	0.33		0.31–0.35	(Hao et al., 2015; Lee et al., 2018; Li and Taghizadeh-Hesary, 2022; Perčić et al., 2021; Song et al., 2017; Zhang et al., 2022a,b)
	Electricity consumption (kWh/TEU-km)	1.30		1.20–1.50	
	LNG consumption (Nm^3/TEU-km)	0.375		0.229–0.536	

Table 6
Share of energy use and emission reduction of vehicles with different energy powers.

Vehicles	Energy type	Share of energy use	Emission reduction factor	Sources
Trucks	Diesel	87.05 %	0 %	(Lee et al., 2018; Song et al., 2017; Zhang et al., 2022b)
	LNG	8.00 %	3 %-13 %	
	Electricity	4.45 %	36 %-43 %	
	Fuel Cell	0.50 %	20 %-45 %	
Trains	Electricity	74.9 %	–	Railway Statistical Bulletin of 2021
	Diesel	25.1 %	–	
Ships	Diesel	98.98 %	0 %	(Fan et al., 2021b, 2021a; Gilbert et al., 2018; Perčić et al., 2021; Sun et al., 2013; Yan et al., 2023)
	LNG	1.00 %	0 %	
	Electric Battery	0.01 %	78.8 %-91.3 %	

are shown in Table 4. Our study combines the research data of transport enterprises in the YRD and the energy consumption data of transit processes from other studies (Guo et al., 2022; He et al., 2015, 2020). The specific energy consumption data are calculated for different hub node operations.

The transport vehicle data in Table 5 encompasses energy consumption data, energy structure ratios, and emission reduction data for various transport vehicles. Data sources include the survey of transport enterprises in the YRD and findings from previous literature.

In this case, the energy emission estimation model utilizes the well-to-wheel (WTW) model, considering the emissions from the energy production process. The emission factor for diesel in the WTW model is 76 gCO_2/MJ, and for natural gas is 69 gCO_2/MJ. As for electricity energy emissions, the data published by the Chinese Ministry of Ecology and Environment from 2012 to 2021 is used. The carbon emission factor is 581 gCO_2/kWh in 2021.

In addition, we consider multiple energy scenarios for ships and trucks. Diesel emissions are set as the benchmark for converting emissions from other fuels using the emission reduction factors shown in Table 6. The carbon emission reduction factors of vehicles under different energy applications are derived from previous studies.

4.3. Scenario design

The scenario design provides the dataset for carbon peaking simulation. Based on the findings of our previous carbon sensitivity analysis for intermodal networks (Guo et al., 2022), we develop two comprehensive scenarios to explore the low-carbon pathways for transportation networks: the network element-based scenario and the energy technology-based scenario.

(1) Network element-based scenario design.

In the network element-based scenario, we focus on two critical factors of transport structuring (TS) and network connectivity (NC) co-defined as bi-control measures. The TS factor represents the share of transport modes shifting within the intermodal network. The NC factor measures the connectivity of networks in different transportation modes, considering factors such as node degree and closeness centrality.

Considering the current policy objectives of government planning, we develop benchmark scenarios for network elements. The benchmark scenario is defined as BTS-BNC, including the development of TS and NC in different periods in the future. Based on the

Table 7
Parameter setting for the bi-control measures.

Scenarios	Parameter	2022–2035	2036–2040	2041–2050
BTS	Annual growth rate of road-to-rail intermodal container freight	15 %	10 %	3 %
	Annual growth rate of road-to-water intermodal container freight	1 %	0.2 %	–
HTS	Annual growth rate of road-to-rail intermodal container freight	16 %	12 %	5 %
	Annual growth rate of road-to-water intermodal container freight	1.6 %	0.5 %	–
LTS	Annual growth rate of road-to-rail intermodal container freight	12 %	8 %	1.5 %
	Annual growth rate of road-to-water intermodal container freight	0.6 %	0.1 %	–
BNC	Average node degree of rail network	30		
	Average node degree of water network	18.18		
	Average closeness centrality of rail network	0.002538		
	Average closeness centrality of water network	0.002796		
HNC	Average node degree of rail network	39		
	Average node degree of water network	28.91		
	Average closeness centrality of rail network	0.002536		
	Average closeness centrality of water network	0.002354		
LNC	Average node degree of rail network	22.09		
	Average node degree of water network	12.18		
	Average closeness centrality of rail network	0.002610		
	Average closeness centrality of water network	0.002935		

Table 8
Network efficiency of bi-control measures crossover scenarios.

Scenarios	Network efficiency mean	2022–2035	2036–2040	2041–2050
BTS- BNC	Railways	9.93	29.98	40.67
	Waterways	37.04	51.07	61.00
	Highways	112.79	124.95	141.62
HTS- HNC	Railways	16.65	56.09	84.44
	Waterways	53.56	75.59	90.13
	Highways	96.50	82.00	76.69
HTS- LNC	Railways	8.07	26.63	39.22
	Waterways	24.54	33.83	40.29
	Highways	129.90	148.34	166.14
LTS- HNC	Railways	9.93	26.54	35.55
	Waterways	45.87	63.69	76.33
	Highways	112.53	127.63	145.69
LTS- LNC	Railways	4.87	12.88	17.08
	Waterways	21.69	29.88	35.79
	Highways	136.79	169.25	198.11

BTS, a high transport structuring scenario (HTS) and a low transport structuring scenario (LTS) are designed, as shown in Table 7. Only the growth rates of rail and water transport are considered since the road transport volume would decrease with the increase of rail and water transport. Based on the BNC, the high network counteractivity scenario (HNC) and low network connectivity scenario (LNC) are developed according to the connection between hub and spoke nodes.

We obtain four different state of network element scenarios by combining the different TS and NC factors. As shown in Table 8, the bi-control measures crossover scenario reflects the different container flow layouts resulting from the changes in transportation structure and network node layouts under the same container freight volume. These changes are integrated into the network efficiency indicators. Network efficiency is determined by the strength of transport structuring and network connectivity. In the HTS-HNC measures, transport structure and network optimization are performed simultaneously and rapidly when the rail and waterway networks are of the highest efficiency. Conversely, both measures are in slow development, and network efficiency is at the lowest level.

(2) Energy technology-based scenario design.

In the energy technology-based scenario, the utilization of clean energy for different equipment of transport and transfer is addressed. A benchmark energy scenario (BE) is developed upon insights from the energy projections of the International Energy Agency (IEA), carbon emission targets, and predictions of transport energy development, as shown in Table 9. The BE scenario considers the current energy mix in China and assumes that fossil energy will dominate as the energy power for truck and ship transportation until 2050. After 2050, alternative energy sources will gradually replace fossil fuels. As for trucks, electricity-related energy may become increasingly prevalent. In the shipping sector, battery-powered container ships would present growing significance in inland waterway transport, aligning with the future river transport plan of the YRD transport sector.

To explore different possibilities, we develop an advanced energy scenario (AE) and a conservative energy scenario (CE) based on the benchmark energy scenario. In the advanced scenario, energy technology experiences significant breakthroughs, and clean energy theriogenology develops faster than the benchmark scenario within 5 years. Conversely, in the conservative scenario, the energy

Table 9
Share of fuel use in different energy scenarios.

Scenarios	Applications	Type	2030	2040	2050	2060	Sources
BE	Trucks	Diesel	76.88 %	49.58 %	22.68 %	4.07 %	(Fan et al., 2021a; Gilbert et al., 2018; Hao et al., 2015, 2012; Perčić et al., 2021; Stéphanie Bouckaert et al., 2021; Sun et al., 2013; Xing et al., 2021; Yan et al., 2023)
		Natural gas	11.51 %	8.46 %	7.66 %	6.92 %	
		Electricity	10.61 %	37.64 %	58.46 %	59.96 %	
		Fuel cell	1.00 %	4.31 %	11.20 %	29.05 %	
	Trains	Electricity	80.62 %	85.71 %	88.33 %	89.96 %	
		Diesel	19.38 %	14.29 %	11.67 %	10.04 %	
	Ships	Diesel	85.15 %	61.51 %	38.87 %	23.16 %	
		LNG	3.08 %	4.92 %	4.93 %	1.36 %	
		Electric Battery	11.77 %	33.57 %	56.20 %	75.48 %	
	Transferring clean energy		11.79 %	47.69 %	85.41 %	94.35 %	
	Electricity emission factor (kg/kWh)		0.5317	0.4331	0.3185	0.1390	
AE	Trucks	Diesel	66.07 %	38.08 %	15.44 %	6.99 %	
		Natural gas	11.87 %	8.05 %	7.28 %	6.92 %	
		Electricity	19.99 %	46.91 %	62.98 %	67.85 %	
		Fuel cell	2.08 %	6.95 %	14.29 %	18.24 %	
	Trains	Electricity	83.36 %	87.08 %	89.18 %	89.96 %	
		Diesel	16.64 %	12.92 %	10.82 %	10.04 %	
	Ships	Diesel	76.26 %	51.66 %	31.23 %	20.55 %	
		LNG	3.89 %	4.92 %	2.59 %	1.36 %	
		Electric Battery	19.85 %	43.41 %	66.18 %	78.09 %	
	Transferring clean energy		11.79 %	47.70 %	93.83 %	98.61 %	
	Electricity emission factor (kg/kWh)		0.4809	0.3716	0.1390	0.1031	
CE	Trucks	Diesel	82.33 %	68.07 %	38.08 %	15.44 %	
		Natural gas	10.40 %	9.87 %	8.05 %	7.28 %	
		Electricity	6.59 %	19.99 %	46.91 %	62.98 %	
		Fuel cell	0.68 %	2.08 %	6.95 %	14.29 %	
	Trains	Electricity	76.85 %	83.36 %	87.08 %	89.18 %	
		Diesel	23.15 %	16.64 %	12.92 %	10.82 %	
	Ships	Diesel	98.01 %	76.26 %	51.66 %	31.23 %	
		LNG	1.65 %	3.89 %	4.92 %	2.59 %	
		Electric Battery	0.34 %	19.85 %	43.41 %	66.18 %	
	Transferring clean energy		10.00 %	25.45 %	41.46 %	67.53 %	
	Electricity emission factor (kg/kWh)		0.5619	0.4809	0.3716	0.2109	

technology development is slower, and clean energy technology lags behind the benchmark scenario by 5 years. The specific parameters and trend changes are shown in Table 9 and Fig. 4.

5. Numerical results and scenario analysis

5.1. Hub identification and freight flow estimation in the YRD-CIN

The intermodal network modeling dataset comprises 20 feature data points from 41 cities and has been verified to be outliers-free. We divided the dataset into 80 % training and 20 % test sets in the random forest model. By setting the number of trees to at least 100 and the maximum depth to 3, the experimental results indicate that the mean cross-validation score is 0.936, with a confidence interval of [0.906, 0.938]. Furthermore, the weighted average precision, recall, and F1 measure 0.90, 0.89, and 0.88, respectively. The model achieved these results with a running time of 2.44 s. Subsequently, we increased the number of trees to 5000 and observed similar results, with the mean cross-validation score and performance metrics remaining consistent. However, the running time increased to 36.61 s. Fig. 5 displays the feature importance rankings as obtained from the RF results. The critical results show that the node connectivity indicators proposed in this study rank higher.

Data from 41 cities are collected, and ten hubs are identified within the YRD-CIN. The radiation hierarchy relationship of hub nodes is established in YRD-CIN, as shown in Fig. 6. The network revolves around key cities such as Shanghai, Nantong, Nanjing, Hangzhou, and Ningbo, with internal coverage connecting cities like Suzhou and Wuxi. This configuration represents the most densely connected intermodal transport network. Also, the network further expands throughout the entire YRD and builds connections with other hub cities.

For the transportation modes of the hub cities, we observe that road transport remains the dominant mode of container transportation in the YRD. It accounts for an average freight volume of 69.1 %. The next is water transport, accounting for 27.12 % of the total volume. The notable development of inland waterway transport is prosperous in coastal city hubs, such as Shanghai, Ningbo, and

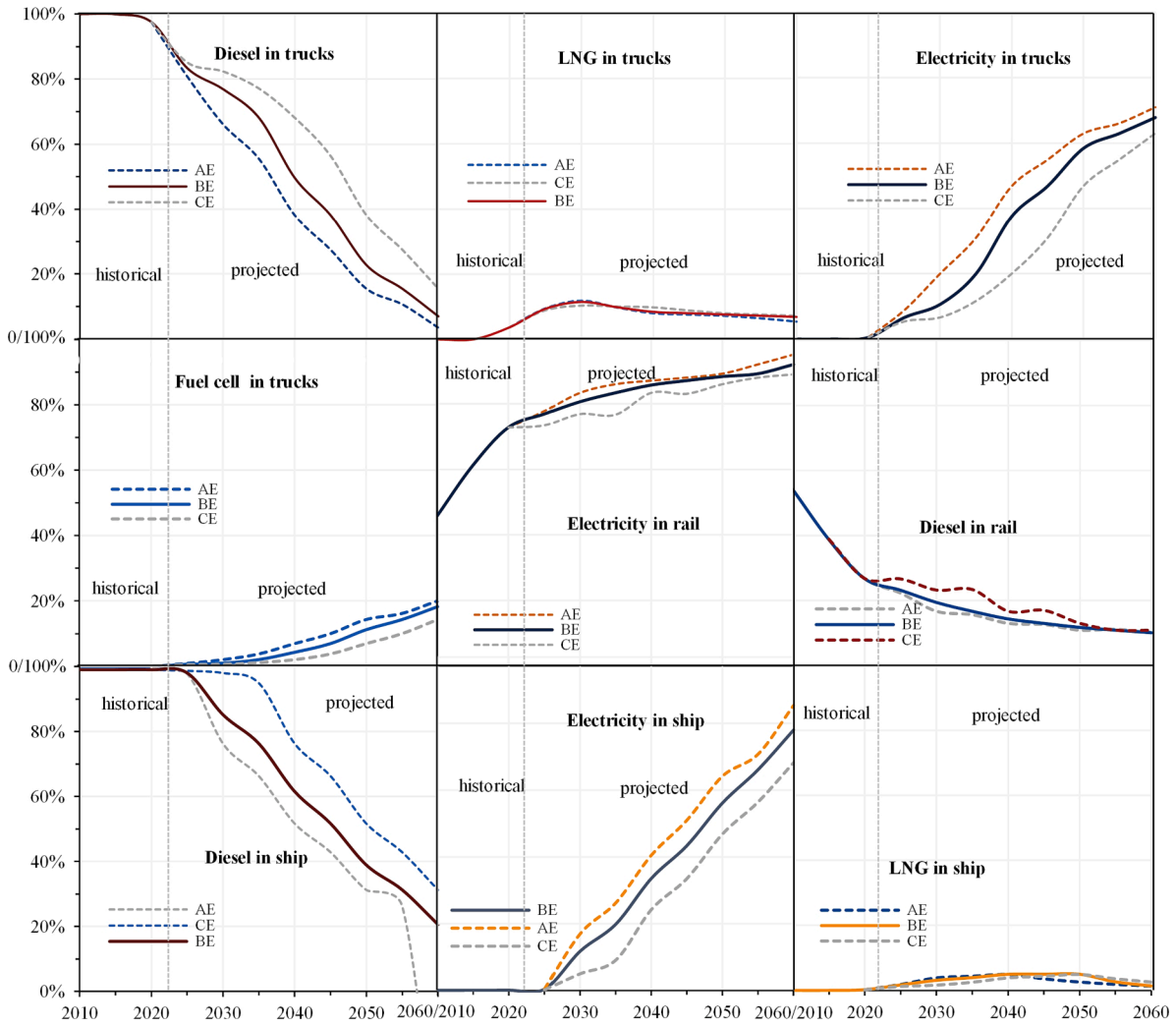


Fig. 4. Fitting curves of energy share assumptions for different transportation modes in 2012–2060.

Lianyungang, and trunk city hubs along the Yangtze River, like Nantong and Suzhou. It may not be surprising that rail transport lags in the YRD-CIN, with an average of only 2.99 % on freight volume average. Accordingly, the average network efficiency for rail, waterway, and highway networks over the past decade is 1.97, 22.36, and 67.63. It is inferred that rail transport has limited participation in the YRD-CIN, and the significant potential for improvement is herein.

Using a freight flow estimation model, the container freight volume of hub nodes is determined within the YRD-CIN from 2012 to 2021, as shown in Fig. 7. Additionally, we obtain information on the flow and direction of freight as presented in Appendix Table A3 for 2021. The container freight volume within the YRD-CIN is consistently growing, with an average annual growth rate of 5.51 %. Among the hubs, Shanghai and Ningbo lead in freight volume, accounting for 63.72 % of the total YRD-CIN, as shown in Table 10.

5.2. *Historical carbon emissions and static carbon peak projections of the YRD-CIN*

With the available data on freight flow at the hub nodes, we estimate the historical carbon emissions of the YRD-CIN. In Fig. 8, the carbon emissions of the YRD-CIN increased over the years. From 2012 to 2021, the emissions grew from 5.13 million tons of carbon emission ($MtCO_2$) to 7.78 $MtCO_2$, with an average annual growth rate of 4.78 %. As for the carbon emission contribution from each hub city, Shanghai ranks first with 40.64 % of the total regional emissions over the past decade, followed by Ningbo with 25.10 %. This is primarily due to the overwhelmingly vibrant container trade activities in these port cities than those of other hubs.

Fig. 9 shows the carbon emissions of different transport modes within the YRD-CIN. Road transport is the primary emission contributor, accounting for an average share of 86.11 % over the past decade. Carbon emissions from inland waterway transport follow with a share of 12.59 %. The Shanghai Port generates the most significant contribution, representing an average share of 66.07 % in the inland waterway transport carbon emission. One of the reasons is that the growth of container waterborne transit in Shanghai is

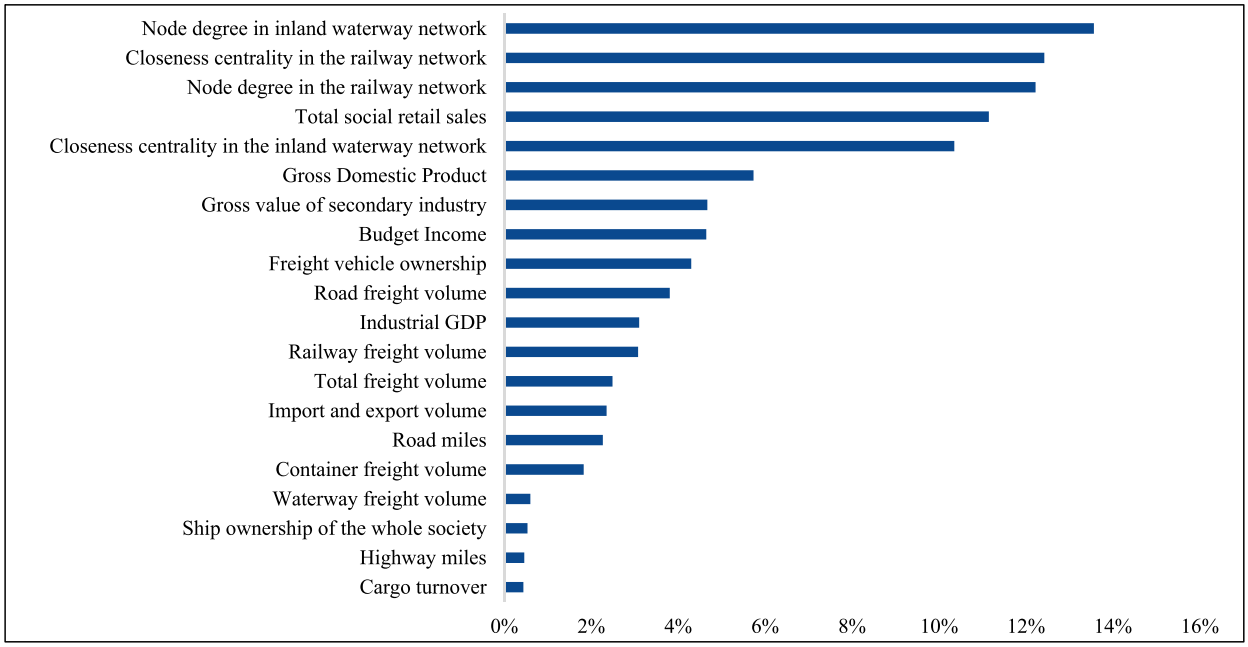


Fig. 5. Feature importance ranking.

remarkably rapid, exceeding 50 % over the past decade. Rail transport features the lowest emission contribution, with an average annual share of only 1.3 %. This can be attributed to the relatively slower development of rail container transport in YRD-CIN.

Fig. 10 shows the changing carbon emission intensity of containers in the YRD-CIN over the past decade. The data indicate a decreasing trend in carbon emission intensities, with an average annual decrease of 0.45 %. The carbon reduction measures implemented during this period are likely effective. From a city perspective, the Shanghai hub has the lowest carbon emission intensity, recording 0.519 kg of CO_2 per kilometer of a container ($kgCO_2$/TEU-km) in 2021. Nantong and Hefei hubs also present significant decreases over the past decade, with reductions of 9.85 % and 9.5 %, respectively. The decrease in Nantong is attributed to the increase in ship transportation volume by 69.6 % over ten years. Hefei shows joint growth in ship and rail transport, with increases of 78.6 % and 190 %, respectively, over the same period. These findings infer that the transport structure significantly impacts carbon reduction efforts.

By identifying benchmark scenario parameters, a static prediction of the future carbon peak state of the YRD-CIN is developed. Fig. 11 illustrates the potential range of carbon peak paths based on static predictions in the case of different confidence intervals. Under the current development trend and scenarios in policy planning, the carbon emissions of the YRD-CIN will peak between 2033 and 2040, setting a peak range of 9.81–11.08 $MtCO_2$ with a 99 % confidence interval. Additionally, the carbon intensity of the YRD-CIN is expected to decrease at an average annual rate of 2.97 % during the peak interval, ranging from 0.426 to 0.352 $kgCO_2$/TEU-km. To gain insight into the changing patterns of the carbon peak, a more detailed scenario analysis is conducted afterward, aiming to identify the decisive factors influencing these patterns.

5.3. Carbon peak simulation in the YRD-CIN

5.3.1. Carbon reduction effect analysis under bi-control measures scenarios

Under the benchmark scenario of BTS-BNC, the simulations of carbon peak time follow a negative binomial distribution with a mean of 2036 and a standard deviation of 3 years. As shown in Fig. 12, the probability of the carbon peak occurring before 2035 in this scenario is 31.88 %, while the probability of it happening before 2030 is 0.24 %. The carbon peak follows a lognormal distribution, with an average value of 10.56 $MtCO_2$ and a standard deviation of 0.12 $MtCO_2$.

Compared with the benchmark scenario, carbon peak time and peak distribution vary over different measures in the crossover scenario of bi-control measures. As shown in Fig. 13, at the highest adjustment level of transport structure and network connectivity (HTS-HNC), the average peaking time is 2034, 2 years earlier than the benchmark scenario. The probability of peaking before 2035 is 65.71 %, and before 2030 is 3.96 %. It is higher than the benchmark scenario by 33.38 % and 3.72 %, respectively. The average value of the peak in this scenario is 9.19 $MtCO_2$, 1.37 $MtCO_2$ lower than the BTS-BNC scenario. At the lowest transport structure and network connectivity level in LTS-LNC, the average peaking time is 2039 and 4 years later than the benchmark scenario. The probability of peaking before 2035 is 4.22 %, and the peak will only occur after 2030. The average value of the peak is 12.23 $MtCO_2$, which is 1.66 $MtCO_2$ higher than that in the BTS-BNC scenario. These demonstrate that the bi-control measures have a significant effect on carbon peak.

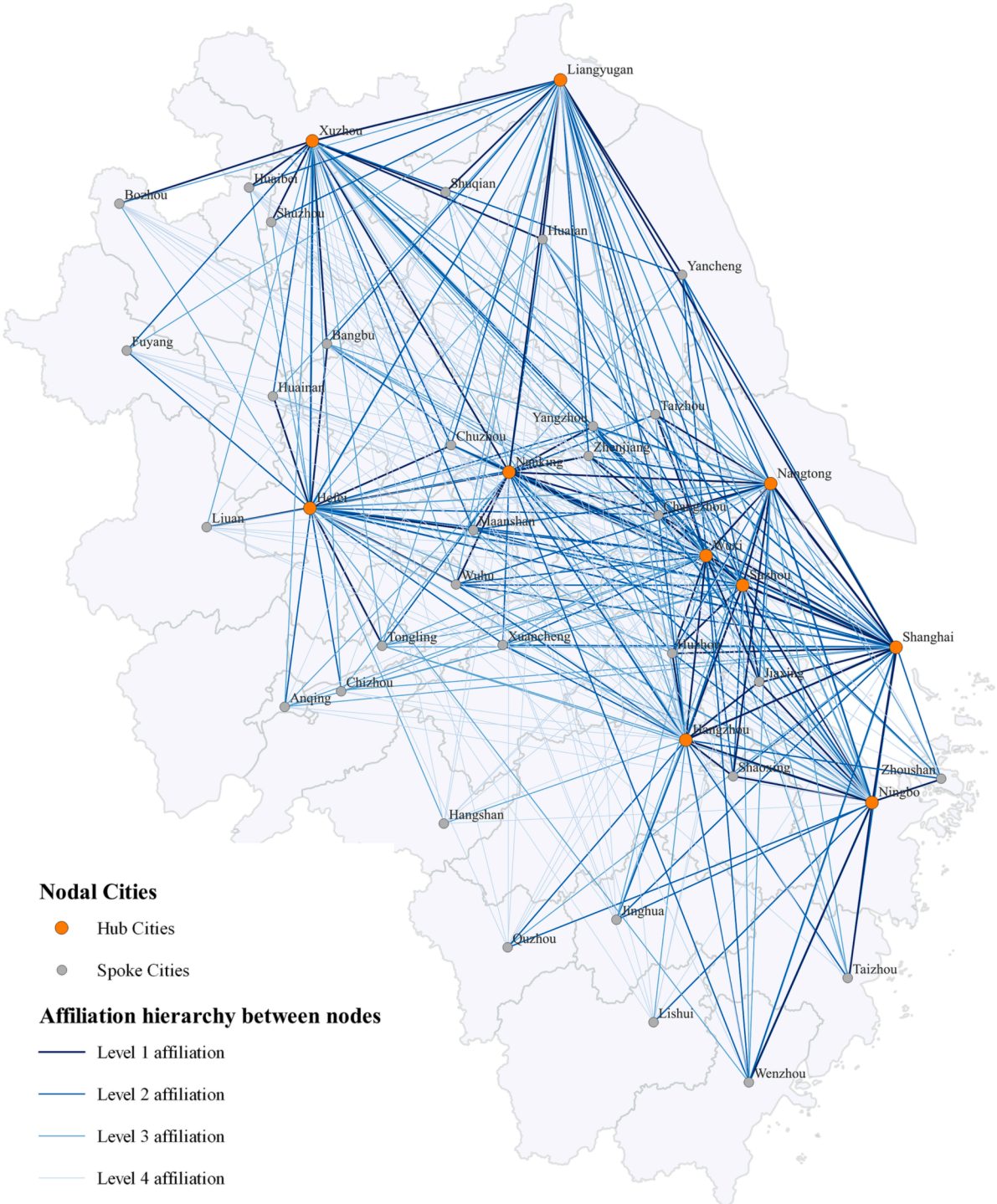


Fig. 6. Hub node service network.

In the BTS-BNC scenario, an earlier peak time occurs than the HTS-LNC measure. The BTS-BNC scenario has a peak value of 0.66 $MtCO_2$, more than HTS-LNC. It is addressed that only the increase of the magnitude of transportation restructuring without considering network connectivity construction is less effective than the synergistic adjustment of both measures. Therefore, in government decision-making, if following the lowest peak path through high levels of bi-control measures is hardly feasible, generating a balanced value of the bi-control measures effectively achieves the optimal level of network peak.

When comparing the crossover scenario of bi-control measures, it is observed that although HTS-LNC peak 3 years earlier, its peak

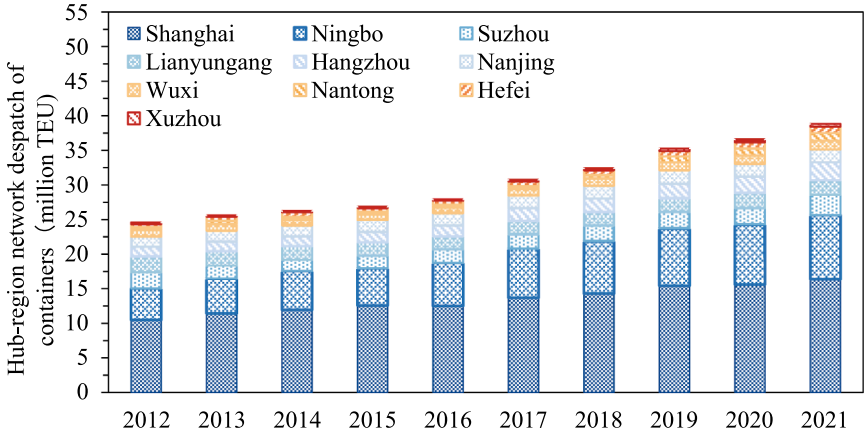


Fig. 7. The container freight volume of hub nodes of the YRD-CIN.

Table 10
The ratio of the three transportation modes of the hub nodes.

Hub nodes	Road	Waterway	Railway	Connectivity	Sources
Average	69.10 %	27.12 %	2.99 %	0.594	Urban Statistical Yearbook for 2012–2021 and calculation results
Shanghai	55.60 %	43.10 %	1.30 %	0.639	
Nanjing	51.75 %	46.13 %	2.12 %	0.669	
Wuxi	85.83 %	13.58 %	0.59 %	0.662	
Xuzhou	76.26 %	10.48 %	13.26 %	0.547	
Suzhou	91.90 %	6.74 %	1.36 %	0.667	
Lianyungang	59.73 %	37.28 %	2.99 %	0.619	
Hangzhou	75.05 %	15.07 %	1.31 %	0.499	
Ningbo	53.52 %	41.43 %	5.02 %	0.643	
Hefei	83.59 %	15.85 %	0.56 %	0.444	
Nantong	57.83 %	41.25 %	0.92 %	0.59	

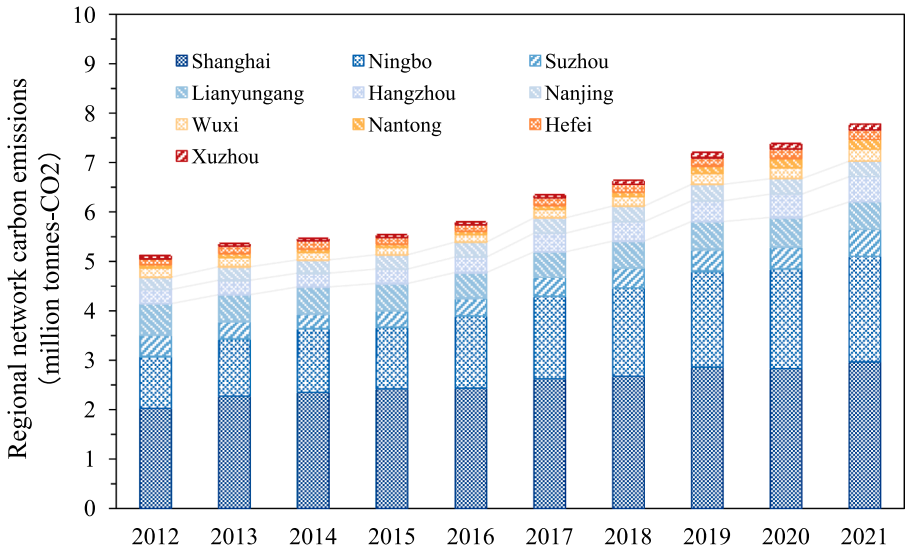


Fig. 8. Carbon emissions in the hub node of YRD-CIN from 2012 to 2021.

value is 0.64 $MtCO_2$ higher than that of LTS-HNC. It indicates that transportation restructuring has a significant impact on the carbon peak time, while network connectivity construction has a significant effect on the peak value. As for the same measure in different adjustment scenarios, when the same transport structure measure is applied to various network connectivity measures, the difference in peak values is more evident than that in peak time. In the HTS scenario, compared to LNC, HNC peak 3 years earlier, with a reduction

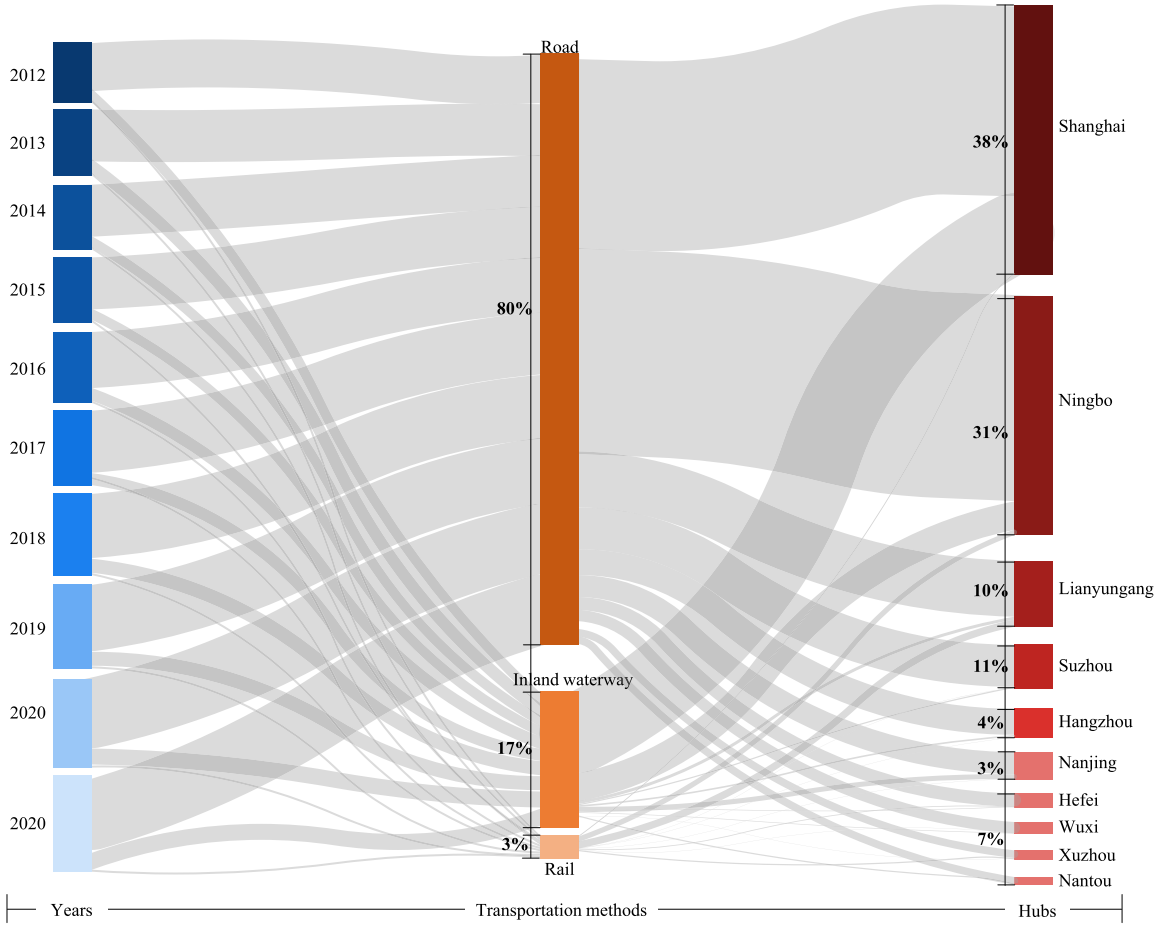


Fig. 9. Percentage distribution of carbon emissions in the YRD-CIN from 2012 to 2021.

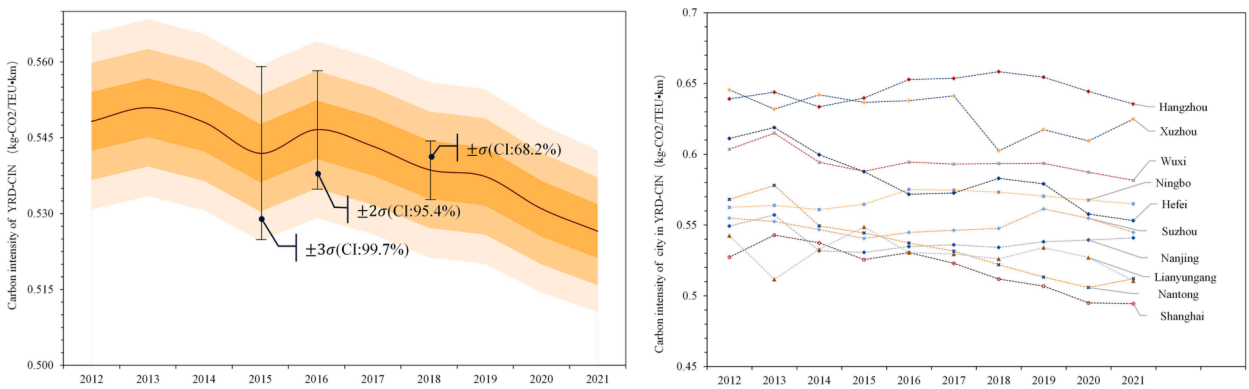


Fig. 10. Interval distribution and changes in hub cities of historical emission intensity in the YRD-CIN.

of 2.03 $MtCO_2$ in the carbon peak. In contrast, in the LTS environment, the peak time of HNC and LNC is very similar, but the peak value of HNC is 1.64 $MtCO_2$ lower. Comparing longitudinally in Fig. 14, when the same network connectivity measures are applied to different transport structure measures, the difference in peak time is more evident than that in peak value. In the HNC scenario, compared to LTS, HTS peak 6 years earlier, with a reduction of 1.39 $MtCO_2$. For the LTS measures, the difference is 3 years in peak time and 1.00 $MtCO_2$ in peak value. It is further concluded that network connectivity measures significantly impact the carbon peak value, while transport structure measures significantly impact the carbon peak time.

Overall, it is shown that the bi-control measures play significant roles in the carbon peak of the YRD-CIN. Transport restructuring

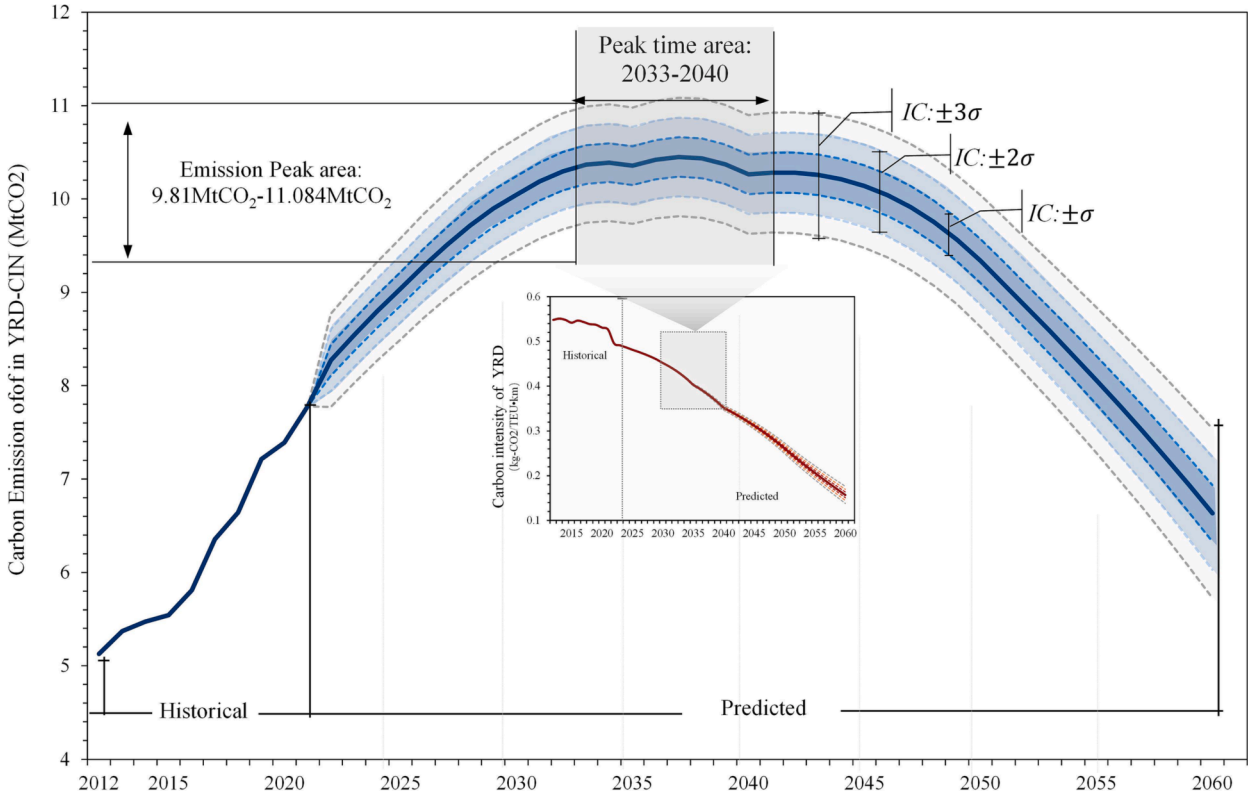


Fig. 11. Static carbon peak trends in the YRD-CIN.

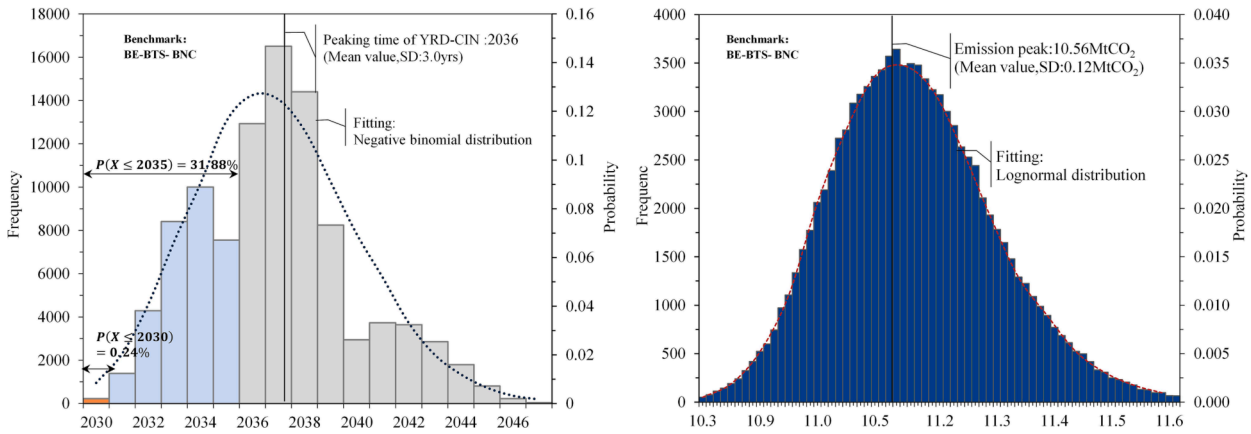


Fig. 12. Regional network carbon peak time and peak distribution.

helps achieve faster carbon peak, while network connectivity helps achieve lower peaks. Furthermore, there is potential for synergistic optimization between the two paths. When making governmental decisions on carbon emission reduction, it is necessary to coordinate the carbon peak time with the peak value and consider the synergy of bi-control measures. Ignoring peak value while pursuing an earlier carbon peak may result in a higher carbon emission burden in the short term.

5.3.2. Reduction effect analysis of bi-control measures under different energy technology scenarios

Based on energy scenarios mentioned in Table 9, the carbon peak scenarios in the YRD-CIN are generated, considering different energy technologies and bi-control measures. Through 100,000 Monte Carlo simulations, we examine the average peak time and values for the five bi-control measures and three energy scenarios, including the benchmark scenario, as presented in Table 11.

Comparing the carbon peak status of the same bi-control measures for different energy technologies leads to insights from Fig. 14. It

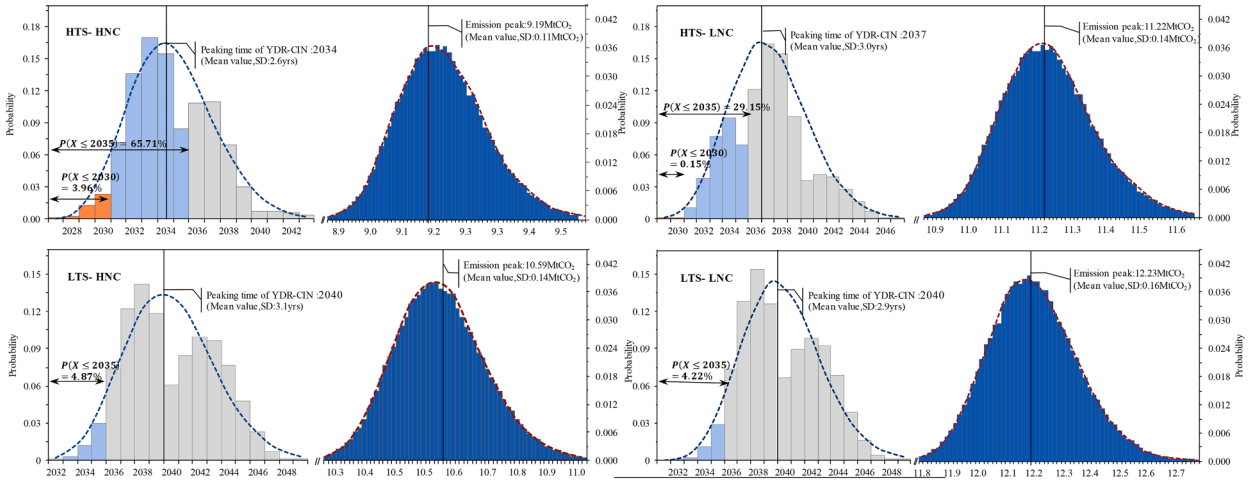


Fig. 13. Carbon peak time and peak probability distribution of the YRD-CIN under the bi-control measures.

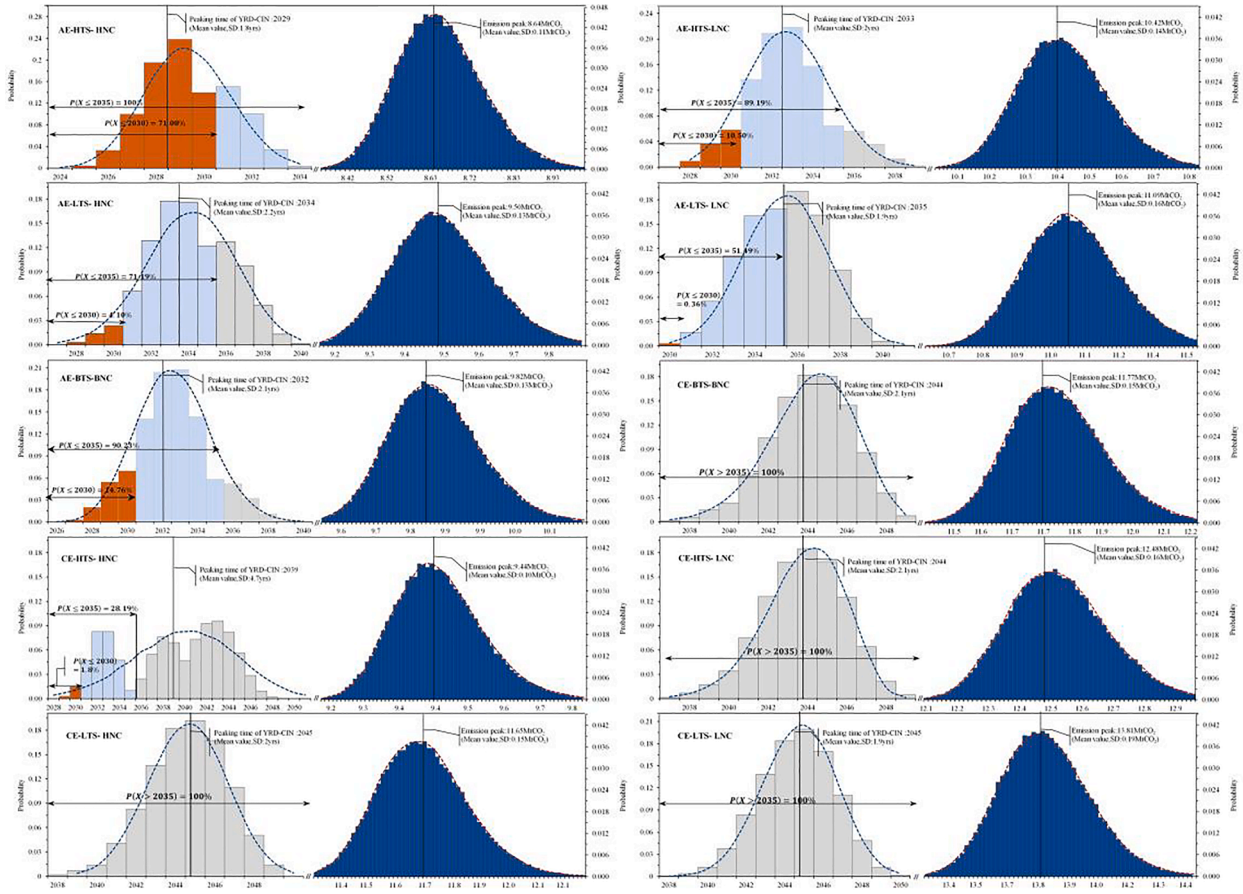


Fig. 14. Probability distribution of carbon peak time and peak value under different energy environments and bi-control measures.

is found that the AE environment is capable of the lowest peak in every bi-control measure, contrary to CE. Furthermore, for different energy technologies, various bi-control measures show different sensitivities. All bi-control measures have the same sensitivity of diffident energy technologies in the carbon peak times. It is found that the difference between the AE and the CE for the five bi-control measures consistently spans over ten years for carbon peak time. Furthermore, in carbon peak value, the sensitivities of bi-control measures for energy technologies show considerable variation. For instance, under the LTS-LNC, the average peak value in the AE

Table 11
Carbon peak time and peak values under different energy environments and bi-control measures.

Scenarios	AE	BE	CE	AE	BE	CE
	Carbon peak times (year)			Carbon peaks ($MtCO_2$)		
HTS-HNC	2029	2034	2039	8.64	9.19	9.44
HTS-LNC	2033	2037	2044	10.42	11.23	12.48
BTS-BNC	2032	2036	2044	9.82	10.56	11.77
LTS-HNC	2034	2040	2045	9.50	10.59	11.66
LTS-LNC	2035	2040	2045	11.09	12.23	13.81

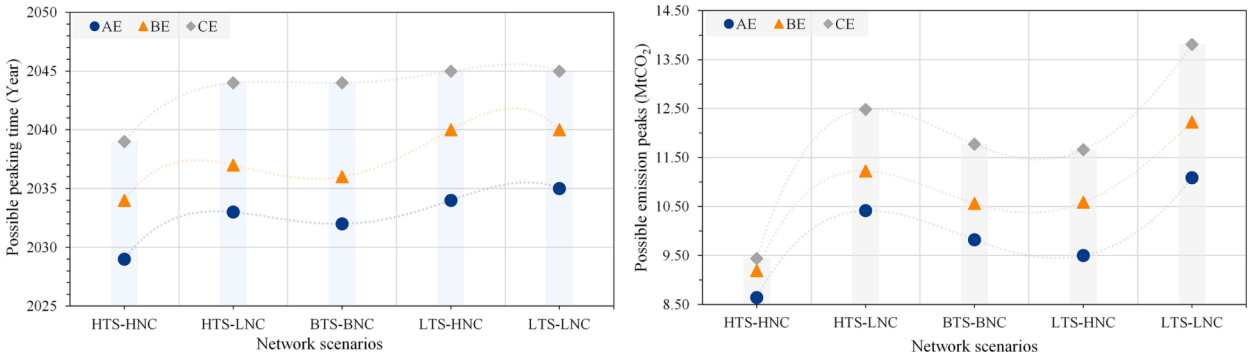


Fig. 15. Comparison of carbon peak time and peak value for the YRD-CIN.

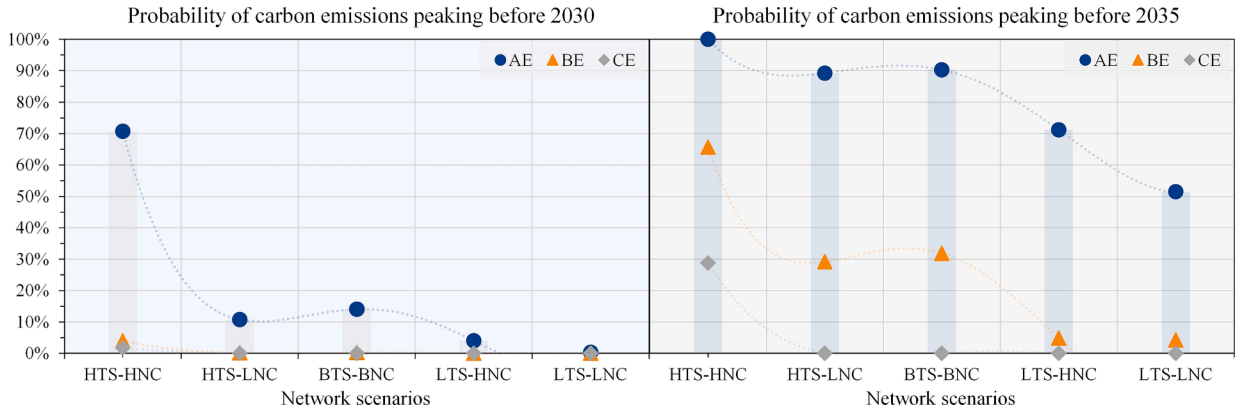


Fig. 16. Probability of achieving different carbon peak targets in the YRD-CIN with different scenarios.

Table 12
Probability of achieving different carbon peak targets in the YRD-CIN.

Scenarios	Carbon peak before 2030			Carbon peak before 2035		
	AE	BE	CE	AE	BE	CE
HTS-HNC	71 %	4 %	1.80 %	100 %	65.71 %	28.82 %
HTS-LNC	11 %	0 %	0 %	89 %	29.15 %	0 %
BUA	14 %	0 %	0 %	90 %	31.88 %	0 %
LTS-HNC	4 %	0 %	0 %	71 %	5 %	0 %
LTS-LNC	0 %	0 %	0 %	51 %	4 %	0 %

scenario is 2.72 $MtCO_2$ lower than in the CE scenario. For the HTS-HNC measure, this gap is only 0.79 $MtCO_2$. It is indicated that while energy technology consistently influences the carbon peak time using different bi-control measures, there are variations in the effectiveness of peak reduction. The sensitivity of energy technologies is much lower for high bi-control measures than for low levels.

When evaluating the gaps in carbon peak at the same energy technology level in Fig. 15, the carbon peak gap between different bi-control measures fluctuates significantly in the CE scenario. HTS-HNC is 4.37 $MtCO_2$ lower than LTS-LNC. In the AE and BE scenarios,

Table A1
Nomenclature.

Nomenclature			
CIN	Container Intermodal Network	NC	Network Connectivity
YRD	Yangtze River Delta	HTS	High Transport Structuring scenario
YRD-CIN	Yangtze River Delta regional Container Intermodal Network	LTS	Low Transport Structuring scenario
GHG	Greenhouse Gas	BTS	Benchmark Transport Structuring scenario
RF	Random Forest	BNC	benchmark Network Connectivity scenario
OD	Origin-Destination	HNC	High Network Counteractivity scenario
GM	Gravity Model	LNC	Low Network Connectivity scenario
LNG	Liquefied Natural Gas	BE	Benchmark Energy
WTW	Well-To-Wheel	AE	Advanced Energy scenario
GDP	Gross Domestic Product	CE	Conservative Energy scenario
TS	Transport Structuring	TEU	Twenty feet Equivalent Unit
L/TEU-km	Liters of fuel consumed by a container transport unit per kilometer	Nm^3/ TEU	Standard cubic meters of LNG consumed by a container transport unit per kilometer
MJ/TEU-km	Megajoules of energy consumed per unit kilometer of transport for a container	kWh/TEU-km	Kilowatt-hours of electrical energy consumed per unit kilometer of transport for a container r
gCO_2	The grams of carbon dioxide emission	$MtCO_2$	The million tons of carbon dioxide emission
$kgCO_2$/TEU-km	kilogram of carbon dioxide emission per kilometer of a container transported		

Table A2
Experimental design.

Step 1: Establishment of network datasets *A*
→ Collecting hub node identification dataset as *Dataset 1*
→ Collecting freight flow estimation dataset as *Dataset 2*
→ Collecting network carbon emission estimation dataset as *Dataset 4*
→ Collecting carbon peak simulation dataset as *Dataset 4*
Step 2: Identifying hubs model based on RF
→ Dataset preprocessing
□Input: *Datasets 1*
- Checking for data outliers
- Standardization
■Output: the sample matrix $\{(X_i, y_i) : i \in [m] := \{1, 2, \ldots, m\}\}$
→ Training RF model
□Input: the sample matrix $\{(X_i, y_i) : i \in [m] := \{1, 2, \ldots, m\}\}$, parameters set
- Dataset segmentation {training data, testing data}
- Parameters set of grid search
- Parameter optimization with cross-validation
■Output: the trained model with Maximum depth and tree number, etc.
→ Identifying hub nodes using the fit model
□Input: randomly sampling from the training data, testing data, selected parameter
■Output: The nodes attributes of $\{Y_i = 1,\ Y_i = 0\}$, routes between the hubs and spokes as ***Output 1***
Step 3: Estimation of the freight flow of routes and hubs
→ Solving Eqn 3.5, 3.6, 3.7, 3.8, 3.9 and 3.10
□Input: Dataset 2 and *Output 1*
■Output: intermodal route freight flows Q_{ij}^t and the freight volume of hubs Q_i^t as ***Output 2***

Step 4: Carbon emission estimation for intermodal networks
→ Solving Eqn 3.11, 3.12 and 3.13
□Input: *Datasets 3,* ***Output 2***
■Output: carbon emission of the intermodal network C_r^t as ***Output 3***
→ Solving Eqn 3.14 and 3.15
□Input: Datasets 3 and ***Output 3***
■Output: Carbon emission intensity for unit containers in the regional network I_r^t.
Step 5: Simulation of carbon peaks and peaking times
→ Scenario design
□Input: Different network connectivity statuses and *Dataset 4*
■Output: Annual growth rate per parameter and network efficiency as ***Output 4***
→ Performing the Monte Carlo simulation
□Input: ***Output 3, Output 4***
- Establishing risk variables and constructing probabilistic
- Random sampling and statistics
- Prediction of carbon emission peaks and peak times
- Probability distribution diagrams
■Output: The probability of carbon peak times T_r^{peak}, carbon values C_r^{peak} and probability distribution diagrams in different scenarios
→ End

Table A3
Estimation of container flow and direction of hub nodes in the YRD-CIN (unit: ten thousand TEU).

Cities	Shanghai	Nanjing	Wuxi	Xuzhou	Suzhou	Nantong	Lianyungang	Hangzhou	Ningbo	Hefei
Shanghai	77.99	9.84	9.77	1.66	33.99	11.07	9.58	19.21	66.85	3.05
Nanjing	54.57	4.49	5.51	1.44	16.34	3.46	7.19	7.4	21.58	5.11
Wuxi	62.95	6.42	2.69	0.93	23.05	3.09	3.72	9.53	16.25	2.09
Xuzhou	18.08	2.96	1.57	1.15	4.99	1.34	9.21	1.82	6.5	1.78
Changzhou	89.16	11.52	6.81	1.46	8.42	7.42	9.3	11.95	22.81	3.29
Suzhou	99.85	8.42	9.21	1.29	14.41	8.84	7.09	16.53	43.68	2.84
Nantong	68.86	3.86	3.03	0.76	19.27	1.85	4.91	5.79	19.98	1.44
Lianyungang	12.5	1.72	0.78	1.17	3.27	1.04	3.7	1.15	6.66	0.61
Huai'an	27.76	5.43	2.7	1.69	4.71	2.23	13.32	1.83	6.65	1.71
Yancheng	45.67	6.06	4.28	1.7	7.47	6.27	16.18	4.01	22.06	2.2
Yangzhou	49.57	11.14	7.17	1.66	10.2	6.26	7.65	7.64	16.41	3.78
Zhenjiang	55.22	9.01	8.91	1.41	11.18	5.22	6.27	8.1	17.08	3.6
Taizhou	54.73	6.16	5.04	1.03	7.6	5.32	3.19	5.15	9.96	1.75
Suqian	20.44	1.85	0.87	0.78	1.62	0.78	10.55	1.4	5.23	0.78
Hangzhou	67.31	4.63	2.77	0.57	20.12	3.24	2.9	13.23	56.36	1.42
Ningbo	46.24	2.67	2.72	0.41	10.59	2.21	3.49	11.15	25.31	0.9
Wenzhou	15.76	1.28	0.75	0.23	3.75	0.9	1.83	2.69	17.64	0.48
Shaoxing	47.42	2.59	2.41	0.48	4.23	1.86	2.66	12.42	74.18	1.12
Huzhou	87.44	5.51	4.04	0.78	8.89	2.85	4.27	9.43	50.66	2.04
Jiaxing	117.78	3.17	8.04	0.57	10.21	3.29	5.46	17.28	62.74	1.78
Jinhua	49.31	3.47	2.8	0.72	6.12	2.23	3.8	14.27	41.49	1.67
Quzhou	26.34	2.07	1.58	0.47	2.83	1.3	2.42	6.77	21.44	1.27
Zhoushan	37.84	1.81	1.01	0.21	3.69	1.6	2.3	6.55	39.57	0.42
Taizhou	42.15	2.25	1.93	0.46	3.8	1.57	3.97	6.95	73.87	1.09
Lishui	22.5	1.65	1.33	0.37	2.2	1.1	2.05	5.41	19.51	0.82
Hefei	21.26	6.38	2.24	1.13	6.9	1.61	3.14	2.87	9.18	2.06
Wuhu	35.48	9.22	4.17	1.2	6.31	2.96	4.44	5.97	17.4	6.66
Maanshan	33.44	6.31	4.16	1.12	5.85	2.94	4.24	4.76	14.55	4.87
Tongling	21.61	4.22	1.6	0.54	2.36	1.29	1.54	2.06	6.23	3.3
Chizhou	20.12	3.45	1.59	0.81	2.42	1.16	2.85	3.57	11.56	2.54
Anqing	20.39	3.33	1.49	0.58	2.44	1.23	1.51	1.83	6	2.8
Xuancheng	31.72	5.27	2.39	0.66	3.38	1.62	3.43	6.74	17.68	2.57
Chuzhou	29.02	6.76	2.9	0.99	4.23	1.99	5.19	4.71	14.75	3.44
Bengbu	21.43	5	1.94	0.97	3.08	1.47	8.04	3.2	11.16	4.16
Huabei	11.9	1.21	0.59	0.85	1.24	0.51	3.04	1.05	3.85	0.84
Huainan	14.42	1.91	0.75	0.98	1.5	0.62	3.88	1.34	4.65	2.29
Suizhou	18.53	3.5	1.59	1.75	3.15	1.29	7.17	2.74	9.94	2.56
Fuyang	19.02	3.22	1.55	1.61	3.1	1.22	5.03	2.92	10.65	3.05
Bozhou	12.15	1.14	0.59	0.82	1.29	0.51	2.3	1.08	4.06	0.86
Liuan	20.18	4.73	2.02	0.94	3.43	1.49	4.26	3.76	13.03	3.29
Huangshan	10.26	1.44	0.83	0.09	1.19	0.63	1.32	2.44	7.07	0.82
Total	1638	187	128	38	295	109	208	259	926	93

this gap is 2.44 $MtCO_2$ and 3.03 $MtCO_2$, respectively. It is highlighted that bi-control measures significantly affect achieving a lower carbon peak in a slow-developing energy technology environment.

The scenarios present significant differences in achieving varied carbon peak time targets, as shown in Fig. 16 and Table 12. To achieve the carbon peak target by 2030, only HTS-HNC under the AE scenario has a higher probability of 71.08 %, followed by AE-BTS-BNC with a probability of only 14.76 %. The probability of achieving this goal under other energy scenarios is scarce. To achieve the carbon peak target by 2030, all five bi-control measures show a higher probability of success in the AE scenario, with an average probability of 80 % and HTS-HNC reaching 100 %. In the BE environment, HTS-HNC has a 65.71 % probability of achieving carbon peak by 2035, followed by BTS-BNC with a probability of 31.88 %. In the CE environment, only the HTS-HNC measure has a 28.82 % probability.

Based on the above analysis, it is evident that attaining the carbon peak before 2030 necessitates meticulous attention to energy technologies and bi-control measures. However, it is easier to achieve the target of carbon peaking before 2035 with synergistic optimization of bi-control measures with appropriate energy technology support. Therefore, it may not be obligatory for the YRD-CIN to strive to achieve the peak by 2030. Instead, it is possible to achieve peak carbon by 2035 through synergistic optimization of transport structure and network.

5.4. Results analysis

It is addressed from the above scenario analysis that:

(1) The adjustment of bi-control measures significantly impacts the timing and magnitude of carbon peak in the network. In the benchmark scenario, a high level of transport structure and network connectivity adjustment leads to earlier and lower carbon

peaks. In the HTS-HNC scenario, the average peak time is 2034, two years earlier than that in the benchmark scenario and six years earlier than the LTS-LNC scenario. The average peak value is 9.19 $MtCO_2$. The probability of peaking before 2030 and 2035 is 65.71 % and 3.96 %, respectively. Furthermore, there is a potential for co-optimization between the dual control measures. Large-scale transportation restructuring contributes to achieving an earlier carbon peak while optimizing network connectivity helps achieve a lower carbon peak value.

(2) Carbon peaking under different bi-control measures has different sensitivities to energy technologies. In a high level of dual control measures, energy technologies have a weak contribution to carbon peaks. In a slow-developing energy technology environment, the dual-control measures significantly affect carbon peak. The peak value difference is 2.72 $MtCO_2$ lower for LTS-LNC and 0.79 $MtCO_2$ for HTS-HNC in AE compared to CE.

(3) Energy technology development influences the effectiveness of bi-control measures in achieving carbon peak targets. Among the 15 scenarios, only AE-HTS-HNC performs the best in achieving the peak carbon target before 2030, with a probability of 71.08 %. In reaching the peak carbon target before 2035, all five bi-control measures in the AE scenario show positive results, whereby HTS-HNC, HTS-LNC, and BTS-BNC in the BE scenario indicate significant possibilities. However, in the CE environment, only HTS-HNC has a probability of 28.82 %, inferring the importance of aligning emission reduction targets with energy technology projections to motivate bi-control measures effectively.

We also develop some practical management insights for transportation sectors from those findings.

(1) When making governmental decisions on carbon reduction, government departments must balance the synergy between the bi-control measures to reach an earlier carbon peak with lower emissions. The transportation sectors should prepare for the shift from highway freight volumes to rail and inland waterways and encourage intermodal transportation services. Meanwhile, transportation resource integration is to be performed to improve the efficiency of expanding existing nodes and channels, adding new nodes and channels, and improving the overall network layout.

(2) The development of carbon measures for intermodal transport networks should consider the broader context of energy technology advancements. This entails making suitable adaptations to the levels of bi-control measure development in line with the pace of energy technology progress. A more lenient approach to dual-control measures would be appropriate in a rapidly evolving energy technology landscape. Conversely, in a slower-moving energy technology environment, a stringent co-optimization of dual-control measures becomes imperative to meet emission reduction goals effectively.

(3) Before setting emission reduction targets, the transportation sectors should assess the energy technology environment and seize energy technology development opportunities. The targets should be determined on different energy technology projections to motivate bi-control measures effectively. If there is a significant breakthrough in the energy technology of vehicles before 2030, a carbon peak before 2030 is highly expected for the regional network. On the other hand, if the energy technology of vehicles develops slowly, the constraint on peak time needs to be released, and an optimized synergy between transportation structure and network layout can assist in achieving optimal carbon peak.

6. Conclusions

Achieving the carbon peak target in the transportation sector requires a clear understanding of the current trend of carbon peaking in the transportation network. It is also essential to evaluate the impacts of various emission reduction policies on facilitating the carbon peak targets. This paper generates solutions for these challenges considering the carbon peak time and peak value of the intermodal transportation network. In this study, a carbon peak prediction scheme for intermodal transport networks is developed by modeling the intermodal network and simulating the carbon peak of this network, including a machine learning-based hub node identification model, a hub node flow estimation model, a hub and spoke intermodal network carbon emission estimation model and a network carbon peak simulation method. In the case study of the YRD-CIN, the historical carbon emission of the network is assessed. Based on current policy planning, 15 scenarios are designed, allowing for adjustments to network elements and innovation in energy technologies. The carbon peaks, peak times, and the factors influencing the intermodal transport network in the YRD region are examined. By comparing with other research in this area, the major contributions of this paper are mainly reflected in two following aspects:

(1) This study expanded the existing research on carbon emission estimation in intermodal transport networks from individual hub networks to the regional network level. The methodology of estimating transport network carbon emissions from the perspective of intermodal transport networks is developed. The models are established on processes of intermodal transport network hub identifications, freight flow estimations, and carbon emission calculations. It provides the theoretical foundation for studying transport network carbon emissions.

(2) This study fills the research gap in existing studies on the carbon peak of intermodal transport networks. A methodology for simulating carbon peaks in intermodal transport networks was developed. The study designed 15 scenarios around intermodal network elements and energy technology elements. These scenarios highlighted the impact of transport network construction within the intermodal network on the carbon peaking trend. Results from this study assist governmental departments in formulating emission reduction policies and assessing the effectiveness of these policies.

Our study presents several potential avenues for future research. First, we prove the differential impact of the bi-control measures

on carbon peak, and that is a synergy between them. Therefore, the synergistic optimization of the bi-control measures would be a valuable research topic. Second, while our study explores the cross-application of machine learning models in intermodal transport, it has certain limitations. Therefore, it is also valuable to explore the deeper integration of machine learning models with transport network research (Liu et al., 2023). In addition, this paper only focuses on three common transport modes. Therefore, future research endeavors should consider examining emerging transportation methods (Qu et al., 2022; Wang and Qu, 2023) that may significantly influence the attainment of carbon peaks within the transport sector.

CRediT authorship contribution statement

Xiaoyan Guo: Conceptualization, Formal analysis, Methodology, Writing – original draft, Formal analysis, Writing – review & editing. **Junliang He:** Conceptualization, Writing – original draft, Resources, Writing – review & editing, Project administration, Funding acquisition. **Hang Yu:** Conceptualization, Writing – original draft, Writing – review & editing, Funding acquisition. **Mei Liu:** Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China [grant number 72072112, 72001135, 72002125]; Shuguang Program of Shanghai Education Development Foundation and Shanghai Municipal Education Commission [grant number 22SG46]; Shanghai Municipal Education Commission [Grant number 2023SKZD16]; Natural Science Foundation of Shanghai [grant number 22ZR1426600]; Shanghai Science and Technology Committee project [grant numbers 21010501800, 22010501900].

Appendix

(SEE Tables A1-A3).

References

Aakko-Saksa, P.T., Lehtoranta, K., Kuittinen, N., Järvinen, A., Jalkanen, J.-P., Johnson, K., Jung, H., Ntziachristos, L., Gagné, S., Takahashi, C., Karjalainen, P., Rönkkö, T., Timonen, H., 2023. Reduction in greenhouse gas and other emissions from ship engines: Current trends and future options. Prog. Energy Combust. Sci. 94, 101055 https://doi.org/https://doi.org/10.1016/j.pecs.2022.101055.

Ampah, J.D., Yusuf, A.A., Afrane, S., Jin, C., Liu, H., 2021. Reviewing two decades of cleaner alternative marine fuels: Towards IMO's decarbonization of the maritime transport sector. J. Clean. Prod. 320 https://doi.org/10.1016/j.jclepro.2021.128871.

Arbues, P., Banos, J.F., 2016. A dynamic approach to road freight flows modeling in Spain. Transportation (amst) 43, 549–564. https://doi.org/10.1007/s11116-015-9590-5.

Asgharian Rezaei, A., Munoz, J., Jalili, M., Khayyam, H., 2023. A machine learning-based approach for vital node identification in complex networks. Expert Syst. Appl. 214, 119086 https://doi.org/https://doi.org/10.1016/j.eswa.2022.119086.

Basallo-Triana, M.J., Vidal-Holguín, C.J., Bravo-Bastidas, J.J., 2021. Planning and design of intermodal hub networks: A literature review. Comput. Oper. Res. 136, 105469 https://doi.org/https://doi.org/10.1016/j.cor.2021.105469.

Bontekoning, Y.M., Macharis, C., Trip, J.J., 2004. Is a new applied transportation research field emerging?—A review of intermodal rail–truck freight transport literature. Transp Res Part A Policy Pract 38, 1–34 https://doi.org/https://doi.org/10.1016/j.tra.2003.06.001.

Stéphanie Bouckaert, Araceli Fernandez Pales, Christophe McGlade, Uwe Remme, Brent Wanner, 2021. Net Zero by 2050 A Roadmap for the Global Energy Sector.

Cao, Z., Yang, Z., Ye, Z., 2022. Research on comprehensive Evaluation and network construction of regional logistics system under "New Land-sea Corridor". Journal of Systems Science 30, 119–124.

Cariou, P., Parola, F., Notteboom, T., 2019. Towards low carbon global supply chains: A multi-trade analysis of CO2 emission reductions in container shipping. Int. J. Prod. Econ. 208, 17–28 https://doi.org/https://doi.org/10.1016/j.ijpe.2018.11.016.

Cheng, L., Chen, X., De Vos, J., Lai, X., Witlox, F., 2019. Applying a random forest method approach to model travel mode choice behavior. Travel Behav. Soc. 14, 1–10. https://doi.org/10.1016/j.tbs.2018.09.002.

Chocholac, J., Hruska, R., Machalik, S., Sommerauerova, D., Krupka, J., 2021. Customized approach to greenhouse gas emissions calculations in railway freight transport. Applied Sciences (switzerland) 11. https://doi.org/10.3390/app11199077.

Commission, E., 2011. Transport 2050: The major challenges, the key measures Why it matters. Belgium, Brussels.

de Miranda Pinto, J.T., Mistage, O., Bilotta, P., Helmers, E., 2018. Road-rail intermodal freight transport as a strategy for climate change mitigation. Environ Dev 25, 100–110 https://doi.org/https://doi.org/10.1016/j.envdev.2017.07.005.

Dong, J., Li, Y., Li, W., Liu, S., 2022. CO2 Emission Reduction Potential of Road Transport to Achieve Carbon Neutrality in China. Sustainability 14. https://doi.org/10.3390/su14095454.

Fan, A., Wang, J., He, Y., Percic, M., Vladimir, N., Yang, L., 2021a. Decarbonising inland ship power system: Alternative solution and assessment method. Energy 226. https://doi.org/10.1016/j.energy.2021.120266.

Fan, A., Wang, J., He, Y., Perčić, M., Vladimir, N., Yang, L., 2021b. Decarbonising inland ship power system: Alternative solution and assessment method. Energy 226, 120266 https://doi.org/https://doi.org/10.1016/j.energy.2021.120266.

Fan, A., Yan, J., Xiong, Y., Shu, Y., Fan, X., Wang, Y., He, Y., Chen, J., 2023. Characteristics of real-world ship energy consumption and emissions based on onboard testing. Mar. Pollut. Bull. 194, 115411 https://doi.org/https://doi.org/10.1016/j.marpolbul.2023.115411.

Franco, V., Kousoulidou, M., Muntean, M., Ntziachristos, L., Hausberger, S., Dilara, P., 2013. Road vehicle emission factors development: A review. Atmos. Environ. 70, 84–97. https://doi.org/10.1016/j.atmosenv.2013.01.006.

Gilbert, P., Walsh, C., Traut, M., Kesieme, U., Pazouki, K., Murphy, A., 2018. Assessment of full life-cycle air emissions of alternative shipping fuels. J. Clean. Prod. 172, 855–866 https://doi.org/https://doi.org/10.1016/j.jclepro.2017.10.165.
Göçmen, E., Erol, R., 2019. Transportation problems for intermodal networks: Mathematical models, exact and heuristic algorithms, and machine learning. Expert Syst. Appl. 135, 374–387 https://doi.org/https://doi.org/10.1016/j.eswa.2019.06.023.
Guo, X., He, J., Lan, M., Yu, H., Yan, W., 2022. Modeling carbon emission estimation for hinterland-based container intermodal network. J. Clean. Prod. 378 https://doi.org/10.1016/j.jclepro.2022.134593.
Hao, H., Wang, H., Ouyang, M., 2012. Fuel consumption and life cycle GHG emissions by China's on-road trucks: Future trends through 2050 and evaluation of mitigation measures. Energy Policy 43, 244–251 https://doi.org/https://doi.org/10.1016/j.enpol.2011.12.061.
Hao, H., Geng, Y., Li, W., Guo, B., 2015. Energy consumption and GHG emissions from China's freight transport sector: Scenarios through 2050. Energy Policy 85, 94–101 https://doi.org/https://doi.org/10.1016/j.enpol.2015.05.016.
Haywood, L., Jakob, M., 2023. The role of the emissions trading scheme 2 in the policy mix to decarbonize road transport in the European Union. Transp Policy (Oxf). https://doi.org/https://doi.org/10.1016/j.tranpol.2023.06.003.
He, J., Huang, Y., Yan, W., Wang, S., 2015. Integrated internal truck, yard crane and quay crane scheduling in a container terminal considering energy consumption. Expert Syst. Appl. 42, 2464–2487 https://doi.org/https://doi.org/10.1016/j.eswa.2014.11.016.
He, J., Tan, C., Yan, W., Huang, W., Liu, M., Yu, H., 2020. Two-stage stochastic programming model for generating container yard template under uncertainty and traffic congestion. Adv. Eng. Inf. 43, 101032 https://doi.org/https://doi.org/10.1016/j.aei.2020.101032.
Heinold, A., Meisel, F., 2018. Emission rates of intermodal rail/road and road-only transportation in Europe: A comprehensive simulation study. Transp Res D Transp Environ 65, 421–437 https://doi.org/https://doi.org/10.1016/j.trd.2018.09.003.
Huang, W., Liu, H., Zhang, Y., Mi, R., Tong, C., Xiao, W., Shuai, B., 2021. Railway dangerous goods transportation system risk identification: Comparisons among SVM, PSO-SVM. GA-SVM and GS-SVM. Appl Soft Comput 109, 107541 https://doi.org/https://doi.org/10.1016/j.asoc.2021.107541.
IPCC, 2006. 2006 IPCC guidelines for national greenhouse gas inventories.
Kalahasthi, L., Holguín-Veras, J., Yushimito, W.F., 2022. A freight origin-destination synthesis model with mode choice. Transp Res E Logist Transp Rev 157, 102595 https://doi.org/https://doi.org/10.1016/j.tre.2021.102595.
Khajavi, H., Rastgoo, A., 2023. Predicting the carbon dioxide emission caused by road transport using a Random Forest (RF) model combined by Meta-Heuristic Algorithms. Sustain. Cities Soc. 93 https://doi.org/10.1016/j.scs.2023.104503.
Kim, N.S., Van Wee, B., 2014. Toward a Better Methodology for Assessing CO2 Emissions for Intermodal and Truck-only Freight Systems: A European Case Study. Int. J. Sustain. Transp. 8, 177–201. https://doi.org/10.1080/15568318.2011.633689.
Kirschstein, T., Meisel, F., 2015. GHG-emission models for assessing the eco-friendliness of road and rail freight transports. Transp. Res. B Methodol. 73, 13–33 https://doi.org/https://doi.org/10.1016/j.trb.2014.12.004.
Kurtuluş, E., Çetin, İ.B., 2020. Analysis of modal shift potential towards intermodal transportation in short-distance inland container transport. Transp Policy (oxf) 89, 24–37 https://doi.org/https://doi.org/10.1016/j.tranpol.2020.01.017.
Latora, V., Marchiori, M., 2001. Efficient Behavior of Small-World Networks. Phys. Rev. Lett. 67 https://doi.org/10.17877/DE290R-11359.
Lee, D.-Y., Elgowainy, A., Kotz, A., Vijayagopal, R., Marcinkoski, J., 2018. Life-cycle implications of hydrogen fuel cell electric vehicle technology for medium- and heavy-duty trucks. J. Power Sources 393, 217–229 https://doi.org/https://doi.org/10.1016/j.jpowsour.2018.05.012.
Levine, B., Nozick, L., Jones, D., 2009. Estimating an origin-destination table for US imports of waterborne containerized freight. Transp Res E Logist Transp Rev 45, 611–626. https://doi.org/10.1016/j.tre.2008.11.001.
Li, Y., Taghizadeh-Hesary, F., 2022. The economic feasibility of green hydrogen and fuel cell electric vehicles for road transport in China. Energy Policy 160, 112703 https://doi.org/https://doi.org/10.1016/j.enpol.2021.112703.
Li, L., Zhang, X., 2020. Integrated optimization of railway freight operation planning and pricing based on carbon emission reduction policies. J. Clean. Prod. 263, 121316 https://doi.org/https://doi.org/10.1016/j.jclepro.2020.121316.
Lin, N., 2019. CO2 emissions mitigation potential of buyer consolidation and rail-based intermodal transport in the China-Europe container supply chains. J. Clean. Prod. 240, 118121 https://doi.org/https://doi.org/10.1016/j.jclepro.2019.118121.
Linton, C., Grant-Muller, S., Gale, W.F., 2015. Approaches and Techniques for Modelling CO2 Emissions from Road Transport. Transp. Rev. 35, 533–553. https://doi.org/10.1080/01441647.2015.1030004.
Liu, Y., Wu, F., Liu, Z., Wang, K., Wang, F., Qu, X., 2023. Can language models be used for real-world urban-delivery route optimization? Innovation 4. https://doi.org/10.1016/j.xinn.2023.100520.
Lü, L., Chen, D., Ren, X.-L., Zhang, Q.-M., Zhang, Y.-C., Zhou, T., 2016. Vital nodes identification in complex networks. Phys. Rep. 650, 1–63 https://doi.org/https://doi.org/10.1016/j.physrep.2016.06.007.
Meng, Q., Wang, X., 2011. Intermodal hub-and-spoke network design: Incorporating multiple stakeholders and multi-type containers. Transp. Res. B Methodol. 45, 724–742 https://doi.org/https://doi.org/10.1016/j.trb.2010.11.002.
Muratori, M., Kunz, T., Hula, A., Freedberg, M., 2023. others. US National Blueprint for Transportation Decarbonization, A Joint Strategy to Transform Transportation.
European Parliamentary, 2021. Sustainable and smart mobility strategy.
Paul, A., Mukherjee, D.P., Das, P., Gangopadhyay, A., Chintha, A.R., Kundu, S., 2018. Improved Random Forest for Classification. IEEE Trans. Image Process. 27, 4012–4024. https://doi.org/10.1109/TIP.2018.2834830.
Perčić, M., Vladimir, N., Fan, A., 2021. Techno-economic assessment of alternative marine fuels for inland shipping in Croatia. Renew. Sustain. Energy Rev. 148, 111363 https://doi.org/https://doi.org/10.1016/j.rser.2021.111363.
Qu, X., Zeng, Z., Wang, K., Wang, S., 2022. Replacing urban trucks via ground–air cooperation. Communications in Transportation Research 2. https://doi.org/10.1016/j.commtr.2022.100080.
Ramalho, M.M., Santos, T.A., 2021. Numerical modeling of air pollutants and greenhouse gases emissions in intermodal transport chains. J Mar Sci Eng 9. https://doi.org/10.3390/jmse9060679.
Randrianarisoa, L.M., Gillen, D., 2022. Reducing emissions in international transport: A supply chain perspective. Transp Res D Transp Environ 102, 103074 https://doi.org/https://doi.org/10.1016/j.trd.2021.103074.
Raparthi, N., Phuleria, H.C., 2022. On-road vehicular emission characterization from the road-tunnel measurements in India: Morphology, emission factors, and sources. Environ. Res. 215 https://doi.org/10.1016/j.envres.2022.114295.
Reul, J., Grube, T., Stolten, D., 2021. Urban transportation at an inflection point: An analysis of potential influencing factors. Transp Res D Transp Environ 92, 102733 https://doi.org/https://doi.org/10.1016/j.trd.2021.102733.
Rudzianskaite-Kvaraciejiene, R., Apanaviciene, R., Gelzinis, A., 2015. Modelling the effectiveness of PPP road infrastructure projects by applying random forests. JOURNAL OF CIVIL ENGINEERING AND MANAGEMENT 21, 290–299. https://doi.org/10.3846/13923730.2014.971129.
Shafiei, E., Stefansson, H., Asgeirsson, E.I., Davidsdottir, B., Raberto, M., 2013. Integrated Agent-based and System Dynamics Modelling for Simulation of Sustainable Mobility. Transp. Rev. 33, 44–70. https://doi.org/10.1080/01441647.2012.745632.
Shen, G., Aydin, S.G., 2014. Origin-destination missing data estimation for freight transportation planning: a gravity model-based regression approach. Transp. Plan. Technol. 37, 505–524. https://doi.org/10.1080/03081060.2014.927665.
Shu, Y., Hu, A., Zheng, Y., Gan, L., Xiao, G., Zhou, C., Song, L., 2023. Evaluation of ship emission intensity and the inaccuracy of exhaust emission estimation model. Ocean Eng. 287, 115723 https://doi.org/https://doi.org/10.1016/j.oceaneng.2023.115723.
Song, H., Ou, X., Yuan, J., Yu, M., Wang, C., 2017. Energy consumption and greenhouse gas emissions of diesel/LNG heavy-duty vehicle fleets in China based on a bottom-up model analysis. Energy 140, 966–978 https://doi.org/https://doi.org/10.1016/j.energy.2017.09.011.
SteadieSeifi, M., Dellaert, N.P., Nuijten, W., Van Woensel, T., Raoufi, R., 2014. Multimodal freight transportation planning: A literature review. Eur. J. Oper. Res. 233, 1–15 https://doi.org/https://doi.org/10.1016/j.ejor.2013.06.055.

Sun, X., Yan, X., Wu, B., Song, X., 2013. Analysis of the operational energy efficiency for inland river ships. Transp Res D Transp Environ 22, 34–39 https://doi.org/https://doi.org/10.1016/j.trd.2013.03.002.
Sun, Y., Yu, N., Huang, B.L., 2022. Green road-rail intermodal routing problem with improved pickup and delivery services integrating truck departure time planning under uncertainty: an interactive fuzzy programming approach. COMPLEX & INTELLIGENT SYSTEMS 8, 1459–1486. https://doi.org/10.1007/s40747-021-00598-1.
Talley, W.K., Ng, M., 2018. Hinterland transport chains: A behavioral examination approach. Transp Res E Logist Transp Rev 113, 94–98 https://doi.org/https://doi.org/10.1016/j.tre.2018.03.001.
Tamin, O.Z., Willumsen, L.G., 1989. Transport demand model estimation from traffic counts. Transportation (amst) 16, 3–26. https://doi.org/10.1007/BF00223044.
Tao, X., Wu, Q., 2021. Energy consumption and CO2 emissions in hinterland container transport. J. Clean. Prod. 279, 123394 https://doi.org/https://doi.org/10.1016/j.jclepro.2020.123394.
The PRC, 2021a. The Action plan to peak carbon dioxide emissions before 2030.
The PRC, 2021b. A modern comprehensive transportation system over the 14th Five-Year Plan (2021-2025).
Tian, P., Mao, B., Tong, R., Zhang, H., Qi, Z., 2023. Analysis of carbon emission level and intensity of China's transportation industry and different transportation modes. Climate Change Research. Https://.
UN, 2021. Sustainable Transport, Sustainable Development, in: Interagency Report for Second Global Sustainable Transport Conference. pp. 1–120.
Wang, Y., Cao, Q., Liu, L., Wu, Y., Liu, H., Gu, Z., Zhu, C., 2022. A review of low and zero carbon fuel technologies: Achieving ship carbon reduction targets. Sustainable Energy Technol. Assess. 54, 102762 https://doi.org/https://doi.org/10.1016/j.seta.2022.102762.
Wang, Y., Han, S., Li, J., Li, B., 2019. Empirical decomposition and forecast of peak carbon emissions of five major transportation modes: Taking the three provinces in Northeast China as examples. Resources Science 41, 1824–1836 https://doi.org/10.18402/resci.2019.10.06.
Wang, L., Peng, C., Shi, W., Zhu, M., 2020a. Carbon dioxide emissions from port container distribution: Spatial characteristics and driving factors. Transp Res D Transp Environ 82. https://doi.org/10.1016/j.trd.2020.102318.
Wang, K., Qu, X., 2023. Urban aerial mobility: Reshaping the future of urban transportation. Innovation. https://doi.org/10.1016/j.xinn.2023.100392.
Wang, L., Zhu, X., 2019. Container Loading Optimization in Rail-Truck Intermodal Terminals Considering Energy Consumption. Sustainability 11. https://doi.org/10.3390/su11082383.
Wen, L., Song, Q., 2022. Simulation study on carbon emission of China's freight system under the target of carbon peaking. Sci. Total Environ. 812, 152600 https://doi.org/https://doi.org/10.1016/j.scitotenv.2021.152600.
Wojahn, O.W., 2001. Airline network structure and the gravity model. Transp Res E Logist Transp Rev 37, 267–279 https://doi.org/https://doi.org/10.1016/S1366-5545(00)00026-0.
Xing, H., Stuart, C., Spence, S., Chen, H., 2021. Alternative fuel options for low carbon maritime transportation: Pathways to 2050. J. Clean. Prod. 297 https://doi.org/10.1016/j.jclepro.2021.126651.
Xiong, S., Yuan, Y., Zhang, C., 2023. Achievement of carbon peak goals in China's road transport—possibilities and pathways. J. Clean. Prod. 388, 135894 https://doi.org/https://doi.org/10.1016/j.jclepro.2023.135894.
Xu, G., Lv, Y., Sun, H., Wu, J., Yang, Z., 2021. Mobility and evaluation of intercity freight CO2 emissions in an urban agglomeration. Transp Res D Transp Environ 91, 102674 https://doi.org/https://doi.org/10.1016/j.trd.2020.102674.
Yan, X., He, Y., Fan, A., 2023. Carbon footprint prediction considering the evolution of alternative fuels and cargo: A case study of Yangtze river ships. Renew. Sustain. Energy Rev. 173, 113068 https://doi.org/https://doi.org/10.1016/j.rser.2022.113068.
Yu, K., Strauss, J., Liu, S., Li, H., Kuang, X., Wu, J., 2021. Effects of railway speed on aviation demand and CO2 emissions in China. Transp Res D Transp Environ 94, 102772 https://doi.org/https://doi.org/10.1016/j.trd.2021.102772.
Zhang, Q., Gu, B., Zhang, H., Ji, Q., 2023. Emission reduction mode of China's provincial transportation sector: Based on "Energy+" carbon efficiency evaluation. Energy Policy 177, 113556 https://doi.org/https://doi.org/10.1016/j.enpol.2023.113556.
Zhang, X., Hao, X., Liu, Y., Wu, R., Shan, X., Li, S., 2022a. Contribution of potential clean trucks in carbon peak pathway of road freight based on scenario analysis: A case study of China. J. Clean. Prod. 379, 134669 https://doi.org/https://doi.org/10.1016/j.jclepro.2022.134669.
Zhang, X., Lin, Z., Crawford, C., Li, S., 2022b. Techno-economic comparison of electrification for heavy-duty trucks in China by 2040. Transp Res D Transp Environ 102, 103152 https://doi.org/https://doi.org/10.1016/j.trd.2021.103152.
Zhou, Q., Du, Z., Liu, J., Liang, J., Jiao, Y., 2021. Factors influencing green ship recycling: A conceptual framework and modeling. J. Clean. Prod. 322 https://doi.org/10.1016/j.jclepro.2021.129155.
Zhou, X., Wang, H., Huang, Z., Bao, Y., Guoqing, Z., Liu, Y., 2022. Identifying spatiotemporal characteristics and driving factors for road traffic CO2 emissions. Sci. Total Environ. 834 https://doi.org/10.1016/j.scitotenv.2022.155270.